

TmFeO₃ Fe–Tm K^- Coupling

Restored derivation: physical J operator, PJP projection, local SU(3) qutrit basis, transported-gauge bond formulas, and the physical cross-peak mechanism (vertices, conservation laws, Liouville pathways), grounded in measured TmFeO₃ parameters

Working note

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1 Purpose and conceptual reset

The goal of this note is to formulate the Fe–Tm coupling in a way that is mathematically well-defined all the way down to the projected qutrit operators. The central object is the K^- coupling, namely the bilinear coupling between an Fe spin and the time-reversal-odd Tm qutrit operators

$$\lambda_j^- = \begin{pmatrix} \lambda_j^2 \\ \lambda_j^5 \\ \lambda_j^7 \end{pmatrix}. \quad (1)$$

The superscript “ $-$ ” means time-reversal odd. The notation j labels the Tm sublattice, not a spatial vector component.

Key point

There are three different mathematical objects that must not be confused:

1. $\mathbf{S}_i$: an Fe spin vector, stored in global crystallographic axes.
2. $\mathbf{J}_j$: the physical Tm angular momentum operator in the full $J = 6$ Hilbert space.
3. λ_j^a : a coordinate operator in the local three-level qutrit basis at Tm sublattice j .

Only $\mathbf{S}$ and the full $\mathbf{J}$ are axial vectors. A Gell-Mann operator λ_j^a is not a spatial vector component. It transforms by the adjoint action induced by the chosen qutrit bases.

The physically supported route is therefore

$$\text{physical } \mathbf{S} \text{ and } \mathbf{J} \longrightarrow H_{\text{Fe-Tm}} = \mathbf{S}^T \mathbf{A} \mathbf{J} \longrightarrow P_j \mathbf{J} P_j \longrightarrow K_{ij}^- \lambda_j^- . \quad (2)$$

One should not start by assuming that the space-group rotation $R_x(\pi)$ acts as a closed onsite SU(3) rotation in one fixed Tm qutrit. That is precisely the dangerous shortcut.

2 Physical system: verified TmFeO₃ parameters

Before any formal manipulation we fix the physical setting and the numbers the model is calibrated against. All values below are experimental properties of bulk TmFeO₃; the simulation parameters sit close to them, and the small residual differences are stated explicitly so that “model” and “measurement” are never silently conflated.

Structure and sites. TmFeO₃ is an orthorhombically distorted perovskite, space group $Pbnm$ (No. 62, a nonstandard setting of $Pnma$), with $Z = 4$ formula units per cell. The Fe³⁺ ions occupy the Wyckoff $4b$ sites, whose site symmetry is $\bar{1}$: *each Fe sits on an inversion centre*, so spatial inversion maps every Fe sublattice to itself. The Tm³⁺ ions occupy the $4c$ sites $(x, y, \frac{1}{4})$, whose site symmetry is the single mirror m (the plane $z = \frac{1}{4}$, i.e. m_z); inversion therefore exchanges Tm sublattices in pairs. This is exactly the Fe/Tm permutation structure used in the Pbnm-ingredients section, and it is dictated by the crystallography, not chosen. In particular, the local mirror $m_z = S_1^{-1}I$ that generates the λ^5, λ^7 sign is the physical $4c$ site mirror.

Magnetic ions. Fe³⁺ is $3d^5$, a spin-only ${}^6S_{5/2}$ ion ($S = 5/2$, $L \approx 0$, $g \approx 2$), represented classically by the unit vector $\mathbf{S}_i$. Tm³⁺ is $4f^{12}$ with ground multiplet 3H_6 ($J = 6$, 13 states). In the $C_s(m)$ crystal field the multiplet splits into 13 *singlets* (a non-Kramers ion in a site without rotational degeneracy has no protected doublets); the qutrit keeps the three lowest, $|E_1\rangle, |E_2\rangle, |E_3\rangle$, and represents them by local SU(3) Gell-Mann operators.

Magnetic order and the Γ_2 ground state. The Fe sublattice orders G-type antiferromagnetically at $T_N \approx 632$ K, with a Dzyaloshinskii–Moriya canting that produces weak ferromagnetism. In Bertaut/Yamaguchi mode notation [5] the relevant collinear-mode multiplets are

$$\Gamma_2 = (F_x, C_y, G_z), \quad \Gamma_4 = (G_x, A_y, F_z). \quad (3)$$

On cooling, TmFeO₃ undergoes a continuous spin reorientation $\Gamma_4 \rightarrow \Gamma_{24} \rightarrow \Gamma_2$ between $T_2 \approx 85$ K and $T_1 \approx 93$ K [3, 6]. Below ~ 85 K the equilibrium is Γ_2 : the main AFM vector $\mathbf{G}$ lies along c (G_z) and the weak ferromagnetic moment $\mathbf{F}$ along a (F_x). This $F_x + G_z$ state is the “baseline” used throughout the numerical sections.

Excitation energies: the two axes of the cross peak. The two THz excitations that define the target cross peak are an Fe magnon and a Tm crystal-field transition:

Excitation	THz	meV	cm ⁻¹	role
Fe quasi-FM magnon	0.28	1.16	9.3	low Fe mode
Fe quasi-AFM magnon	0.86	3.56	28.7	pumped q_{AFM} (ω_τ axis)
Tm CEF E_{12} (R_1)	0.52	2.15	17.3	detected coherence (ω_t axis)
Tm CEF E_{23} (R_2)	0.61	2.52	20.3	sets Δ_{13} with E_{12}
Tm CEF E_{13} (R_3)	1.15	4.76	38.4	third qutrit level

Conversion factor: $1 \text{ THz} = 4.136 \text{ meV} = 33.36 \text{ cm}^{-1} = 47.99 \text{ K}$. The three CEF lines are mutually consistent with a three-singlet qutrit, $E_{12} + E_{23} \approx E_{13}$ ($0.52 + 0.61 \approx 1.13 \text{ THz}$), so the onsite splittings in $H_{\text{Tm},0}$ are $\Delta_{12} \approx 0.52$ and $\Delta_{13} \approx 1.13 \text{ THz}$. The simulation uses $q_{\text{AFM}} \approx 0.90$ and $E_{12} \approx 0.50 \text{ THz}$, within a few percent of the measured 0.86 and 0.52 THz [3, 4]. The cross-peak question is whether a coherence on the $\omega_\tau = q_{\text{AFM}}$ axis can be converted into a coherence on the $\omega_t = E_{12}$ axis; the two differ by $\Delta \approx 0.34 \text{ THz}$, a number we return to under energy conservation in section 16.

3 Pbnm ingredients

We use the generators

$$S_1 : (x, y, z) \mapsto (-x, -y, z + \frac{1}{2}), \quad (4)$$

$$S_2 : (x, y, z) \mapsto (x + \frac{1}{2}, -y + \frac{1}{2}, -z), \quad (5)$$

$$I : (x, y, z) \mapsto (-x, -y, -z). \quad (6)$$

For axial vectors, the corresponding spin-space matrices are

$$R_{S_1} = \text{diag}(-1, -1, +1) \equiv R_z, \quad R_{S_2} = \text{diag}(+1, -1, -1) \equiv R_x, \quad R_I = \mathbf{1}. \quad (7)$$

Their product gives

$$R_{S_1 S_2} = R_y = \text{diag}(-1, +1, -1). \quad (8)$$

The sublattice permutations are

Operation	Fe sublattice action	Tm sublattice action
S_1	(03)(12)	(03)(12)
S_2	(01)(23)	(01)(23)
$S_1 S_2$	(02)(13)	(02)(13)
I	identity	(03)(12)

Thus inversion fixes each Fe sublattice but pairs $\text{Tm}_0 \leftrightarrow \text{Tm}_3$ and $\text{Tm}_1 \leftrightarrow \text{Tm}_2$.

4 Full physical Fe–Tm exchange before projection

Let a Fe–Tm bond be denoted

$$b = (i, j; \mathbf{n}), \quad (9)$$

where i is an Fe sublattice, j is a Tm sublattice, and $\mathbf{n}$ is the Tm-cell offset relative to the Fe cell. Before any qutrit truncation, the most general bilinear physical exchange is

$$H_b^{\text{phys}} = \mathbf{S}_i^T A_b \mathbf{J}_j. \quad (10)$$

Here A_b is a real 3×3 tensor with rows labelled by Fe spin components and columns labelled by Tm angular-momentum components:

$$A_b = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}_b. \quad (11)$$

Under a space-group operation g , the physical axial vectors transform as

$$\mathbf{S}_i \mapsto R_g \mathbf{S}_{\sigma_g^{\text{Fe}}(i)}, \quad \mathbf{J}_j \mapsto R_g \mathbf{J}_{\sigma_g^{\text{Tm}}(j)}. \quad (12)$$

Therefore invariance of eq. (10) requires

$$A_{gb} = R_g^T A_b R_g. \quad (13)$$

This equation is safe because it is written in the full physical Hilbert space. No $SU(3)$ truncation has been used.

Warning

The statement that $\mathbf{J}$ is an axial vector is a statement about the full $J = 6$ angular-momentum operator. It is not a statement that $(\lambda^2, \lambda^5, \lambda^7)$ form a spatial vector.

5 Local qutrit bases and the PJP projection

At each Tm sublattice j , define the local qutrit projector

$$P_j = \sum_{n=1}^3 |E_{n,j}\rangle \langle E_{n,j}|. \quad (14)$$

The local qutrit basis is

$$\mathcal{B}_j = \{|E_{1,j}\rangle, |E_{2,j}\rangle, |E_{3,j}\rangle\}. \quad (15)$$

The Gell-Mann operator at that Tm site is defined by

$$\lambda_j^a = \sum_{m,n=1}^3 |E_{m,j}\rangle (\lambda^a)_{mn} \langle E_{n,j}|. \quad (16)$$

Thus λ_j^a is a local operator coordinate in the Tm qutrit. It is not a component of a spatial vector.

The physical projected angular momentum is

$$\tilde{J}_{j,\alpha} = P_j J_\alpha P_j. \quad (17)$$

In the local qutrit basis it is expanded as

$$P_j J_\alpha P_j = \sum_{a \in \{2,5,7\}} \mu_{j,\alpha a} \lambda_j^a. \quad (18)$$

Lemma: why only $\lambda^2, \lambda^5, \lambda^7$ appear

The restriction of (18) to the three *antisymmetric* Gell-Mann matrices is exact, not a small-multipole approximation, and it rests on one physical fact specific to this system. Tm^{3+} is a non-Kramers $4f^{12}$ ion at a site of symmetry m , so its CEF eigenstates are nondegenerate singlets and can be chosen *real* (time-reversal symmetric). In any real basis the angular-momentum operator J_α is purely imaginary and antisymmetric, hence $\langle E_m | J_\alpha | E_n \rangle$ is imaginary and antisymmetric. The only imaginary-antisymmetric Gell-Mann matrices are $\lambda^2, \lambda^5, \lambda^7$, so $P_j J_\alpha P_j$ has *no* component on $\lambda^{1,3,4,6,8}$. The time-odd (λ^-) versus time-even (λ^+) split used throughout this note is therefore a rigorous consequence of time reversal acting on a real singlet basis, not a modelling choice.

For a reference local basis, the projected moment matrix is

$$\mu_0 = \begin{pmatrix} 0 & 2.3915 & 0.9128 \\ 0 & -2.7866 & 0.4655 \\ 5.264 & 0 & 0 \end{pmatrix}, \quad \text{columns } (2, 5, 7). \quad (19)$$

Equivalently,

$$P_0 J_x P_0 = 2.3915\lambda_0^5 + 0.9128\lambda_0^7, \quad (20)$$

$$P_0 J_y P_0 = -2.7866\lambda_0^5 + 0.4655\lambda_0^7, \quad (21)$$

$$P_0 J_z P_0 = 5.264\lambda_0^2. \quad (22)$$

The physical magnetic moment is

$$\mathbf{m}_j^{\text{Tm}} = -g_J \mu_B P_j \mathbf{J} P_j. \quad (23)$$

Thus a lab magnetic field couples to the projected moment as

$$H_{B,\text{Tm}}(t) = -g_J \mu_B \sum_{j,\alpha,a} B_\alpha(t) \mu_{j,\alpha a} \lambda_j^a. \quad (24)$$

The coefficient $\mu_{j,\alpha a}$ carries all local-basis information.

6 The actual SU(3) symmetry action: W_g and D_g

Let g map Tm sublattice j to $j' = \sigma_g^{\text{Tm}}(j)$. The exact qutrit-basis map is

$$[W_g^{(j)}]_{mn} = \langle E_{m,j'} | U_g | E_{n,j} \rangle. \quad (25)$$

It acts on qutrit states as

$$U_g | E_{n,j} \rangle = \sum_m | E_{m,j'} \rangle [W_g^{(j)}]_{mn}. \quad (26)$$

It induces an adjoint action on qutrit operators:

$$U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{j'}^b, \quad (27)$$

where

$$\left[D_g^{(j)} \right]_{ba} = \frac{1}{2} \text{Tr} \left[\lambda^b W_g^{(j)} \lambda^a W_g^{(j)\dagger} \right]. \quad (28)$$

Key point

$W_g^{(j)}$ acts on qutrit states. $D_g^{(j)}$ acts on qutrit operators. R_g acts on physical spatial vector components. These are different matrices in different spaces.

The projected moment covariance is

$$U_g (P_j J_\alpha P_j) U_g^\dagger = \sum_\beta R_{g,\alpha\beta} P_{j'} J_\beta P_{j'}. \quad (29)$$

Substituting eqs. (18) and (27) gives

$$\sum_a \mu_{j,\alpha a} \left[D_g^{(j)} \right]_{ba} = \sum_\beta R_{g,\alpha\beta} \mu_{j',\beta b}. \quad (30)$$

In matrix notation, if D is defined by $\lambda^a \mapsto D_{ba} \lambda^b$, then

$$\mu_{j'} = R_g^T \mu_j D_g^{(j)T}. \quad (31)$$

This is the more theoretically supported relation. It replaces the unsafe shortcut $\lambda_{\text{global}} = \mu^{-1} R \mu \lambda_{\text{local}}$.

Warning

The matrix $\mu^{-1}R_x\mu$ is generally not the adjoint representation of a unitary qutrit rotation. The operation S_2 maps one Tm sublattice to another; it need not act as a closed onsite operation in one fixed qutrit. The remedy is to work with $W_g^{(j)}$ and $D_g^{(j)}$, or to choose transported bases that make selected $D_g^{(j)}$ trivial by construction.

7 Transported gauge: a practical but explicit convention

Choose Tm_0 as the reference qutrit. Define

$$\mathcal{B}_0 = \{|E_{1,0}\rangle, |E_{2,0}\rangle, |E_{3,0}\rangle\}, \quad (32)$$

$$\mathcal{B}_1 = U_{S_2}\mathcal{B}_0, \quad (33)$$

$$\mathcal{B}_2 = U_{S_1S_2}\mathcal{B}_0, \quad (34)$$

$$\mathcal{B}_3 = U_{S_1}\mathcal{B}_0. \quad (35)$$

This is the transported gauge. In this gauge,

$$D_{S_2}^{(0)} = D_{S_1S_2}^{(0)} = D_{S_1}^{(0)} = \mathbf{1}. \quad (36)$$

After restricting to the corresponding transported maps, the proper-coset qutrit action is therefore absorbed into the basis definition. This does not mean that R_x acts trivially on the qutrit. It means we chose the target qutrit basis to be the image of the source qutrit basis.

The physical global projected moment on site j is then

$$P_j J_\alpha^{\text{global}} P_j = \sum_{\beta,a} (R_j)_{\alpha\beta} \mu_{0,\beta a} \lambda_j^a, \quad (37)$$

where

$$R_0 = \text{diag}(+, +, +), \quad R_1 = \text{diag}(+, -, -), \quad R_2 = \text{diag}(-, +, -), \quad R_3 = \text{diag}(-, -, +). \quad (38)$$

This is how a global lab field talks to local Tm λ_j operators in transported gauge:

$$H_{B,\text{Tm}}(t) = -g_J \mu_B \sum_{j,a} \left[\sum_{\alpha,\beta} B_\alpha(t) (R_j)_{\alpha\beta} \mu_{0,\beta a} \right] \lambda_j^a. \quad (39)$$

8 Local mirror and the origin of the λ^5, λ^7 sign

Two operations map Tm_0 to Tm_3 : S_1 and I . In transported gauge the Tm_3 basis is defined with S_1 . Therefore the inversion map differs from the transported map by the local operation

$$h = S_1^{-1}I. \quad (40)$$

Using the coordinate maps,

$$h : (x, y, z) \mapsto (x, y, -z + \frac{1}{2}), \quad (41)$$

which is a local mirror m_z up to a translation.

For a J_z basis state, the mirror can be written as $m_z = IC_{2z}$. Since C_{2z} acts as $e^{-i\pi J_z}$ and the $4f^{12}$ parity is common to all states,

$$U(m_z)|J, m\rangle = (-1)^m |J, m\rangle \quad (42)$$

after dropping the common parity phase. The first two CEF states live in the even- m sector, while the third lives in the odd- m sector. Hence

$$U(m_z)|E_1\rangle = +|E_1\rangle, \quad U(m_z)|E_2\rangle = +|E_2\rangle, \quad U(m_z)|E_3\rangle = -|E_3\rangle. \quad (43)$$

Thus in the qutrit basis

$$W_{m_z} = \text{diag}(+1, +1, -1). \quad (44)$$

Conjugating the Gell-Mann operators gives

$$\lambda^2 \sim -i|E_1\rangle\langle E_2| + i|E_2\rangle\langle E_1| \Rightarrow +\lambda^2, \quad (45)$$

$$\lambda^5 \sim -i|E_1\rangle\langle E_3| + i|E_3\rangle\langle E_1| \Rightarrow -\lambda^5, \quad (46)$$

$$\lambda^7 \sim -i|E_2\rangle\langle E_3| + i|E_3\rangle\langle E_2| \Rightarrow -\lambda^7. \quad (47)$$

Therefore

$$\boxed{M_- = \text{diag}(+1, -1, -1) \quad \text{on} \quad (\lambda^2, \lambda^5, \lambda^7).} \quad (48)$$

9 From physical exchange to projected K^-

Project eq. (10) to the Tm qutrit:

$$H_b^{\text{phys}} = \sum_{\alpha\beta} S_{i,\alpha}(A_b)_{\alpha\beta} P_j J_\beta P_j \quad (49)$$

$$= \sum_{\alpha,a} S_{i,\alpha} \left[\sum_{\beta} (A_b)_{\alpha\beta} \mu_{j,\beta a} \right] \lambda_j^a. \quad (50)$$

Thus

$$\boxed{K_{b,\alpha a}^- = \sum_{\beta} (A_b)_{\alpha\beta} \mu_{j,\beta a}, \quad a \in \{2, 5, 7\}.} \quad (51)$$

In matrix form,

$$\boxed{K_b^- = A_b \mu_j.} \quad (52)$$

The fully gauge-covariant symmetry rule follows by applying both the Fe rotation and the local qutrit adjoint action:

$$H_b^{K^-} = \mathbf{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \quad (53)$$

Under g ,

$$\mathbf{S}_i \mapsto R_g \mathbf{S}_{\sigma_g^{\text{Fe}}(i)}, \quad \lambda_j^a \mapsto \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{\sigma_g^{\text{Tm}}(j)}^b. \quad (54)$$

Therefore

$$\boxed{K_{gb,\beta b}^- = \sum_{\alpha,a} R_{g,\alpha\beta} K_{b,\alpha a}^- \left[D_g^{(j)} \right]_{ba}.} \quad (55)$$

This index equation is the safest form. In matrix notation, with the same D_{ba} convention,

$$\boxed{K_{gb}^- = R_g^T K_b^- D_g^{(j)T}.} \quad (56)$$

Implementation prescription

Equation (55) is the implementation anchor. The Fe row index is transformed by the spatial axial-vector matrix R_g . The Tm λ column index is transformed by the qutrit adjoint matrix $D_g^{(j)}$. In transported gauge many proper-coset D matrices are absorbed into the basis, but

the Fe row signs remain if Fe spins are stored in global axes.

10 One K^- orbit in transported gauge

Take a reference bond with coefficient

$$K_0^- = \begin{pmatrix} k_{x2} & k_{x5} & k_{x7} \\ k_{y2} & k_{y5} & k_{y7} \\ k_{z2} & k_{z5} & k_{z7} \end{pmatrix} = (\mathbf{k}_2 \quad \mathbf{k}_5 \quad \mathbf{k}_7). \quad (57)$$

For the direct proper-coset partners in transported gauge,

$$K_{ii}^- = R_i K_0^-. \quad (58)$$

For the inversion-related partners,

$$K_{i,3-i}^- = R_i K_0^- M_-. \quad (59)$$

Thus the orbit Hamiltonian is

$$H_{K^-}^{\text{orbit}} = \sum_{i=0}^3 \mathbf{S}_i^T R_i [\mathbf{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_7(\lambda_i^7 - \lambda_{3-i}^7)]. \quad (60)$$

This equation is often the most useful way to read the selection rule:

$$\lambda^2 \text{ couples to the inversion-even Tm combination,} \quad (61)$$

$$\lambda^5, \lambda^7 \text{ couple to inversion-odd Tm combinations.} \quad (62)$$

It does not mean that the mirror-odd terms are identically zero. They vanish only after projecting onto an inversion-even Tm sector.

10.1 All eight coefficients of one orbit

Bond	Generating operation	Coefficient	Meaning
(Fe ₀ , Tm ₀)	E	$K_{00}^- = K_0^-$	reference
(Fe ₁ , Tm ₁)	S_2	$K_{11}^- = R_x K_0^-$	Fe y, z rows flip
(Fe ₂ , Tm ₂)	$S_1 S_2$	$K_{22}^- = R_y K_0^-$	Fe x, z rows flip
(Fe ₃ , Tm ₃)	S_1	$K_{33}^- = R_z K_0^-$	Fe x, y rows flip
(Fe ₀ , Tm ₃)	I	$K_{03}^- = K_0^- M_-$	λ^5, λ^7 columns flip
(Fe ₁ , Tm ₂)	$S_2 I$	$K_{12}^- = R_x K_0^- M_-$	row flip + odd columns
(Fe ₂ , Tm ₁)	$S_1 S_2 I$	$K_{21}^- = R_y K_0^- M_-$	row flip + odd columns
(Fe ₃ , Tm ₀)	$S_1 I$	$K_{30}^- = R_z K_0^- M_-$	row flip + odd columns

11 Fixed-Fe and fixed-Tm viewpoints

For a fixed Fe sublattice i , the mirror-odd part is

$$H_i^{5,7} = \mathbf{S}_i^T R_i \mathbf{k}_5 (\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{S}_i^T R_i \mathbf{k}_7 (\lambda_i^7 - \lambda_{3-i}^7). \quad (63)$$

So the cancellation is in the Tm sector: if $\lambda_i^{5,7} = \lambda_{3-i}^{5,7}$, then this fixed-Fe contribution vanishes.

For a fixed Tm site j , collect the λ_j^5 terms:

$$H_{\lambda_j^5} = (\mathbf{S}_j^T R_j \mathbf{k}_5 - \mathbf{S}_{3-j}^T R_{3-j} \mathbf{k}_5) \lambda_j^5. \quad (64)$$

Similarly,

$$H_{\lambda_j^7} = (\mathbf{S}_j^T R_j \mathbf{k}_7 - \mathbf{S}_{3-j}^T R_{3-j} \mathbf{k}_7) \lambda_j^7. \quad (65)$$

This is not $\mathbf{S}_j - \mathbf{S}_{I(j)}$, because inversion fixes the Fe sublattice. The index $3 - j$ enters because it is the Fe partner of the other member of this particular bond orbit.

12 Including the Fe and Tm Hamiltonians

A minimal Fe Hamiltonian in global axes is

$$H_{\text{Fe}} = \sum_{\langle mn \rangle_1} [J_{1xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{1z} S_m^z S_n^z] \quad (66)$$

$$+ \sum_{\langle mn \rangle_2} [J_{2xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{2z} S_m^z S_n^z] \quad (67)$$

$$+ \sum_m [D_x (S_m^x)^2 + D_y (S_m^y)^2 + D_z (S_m^z)^2]. \quad (68)$$

The onsite Tm qutrit Hamiltonian with $E_1 = 0$, $E_2 = \Delta_{12}$, $E_3 = \Delta_{13}$ is

$$H_{\text{Tm},0} = \sum_j \left[-\frac{\Delta_{12}}{2} \lambda_j^3 + \frac{\Delta_{12} - 2\Delta_{13}}{2\sqrt{3}} \lambda_j^8 \right] + \text{constant}. \quad (69)$$

The full leading model restricted to K^- is

$$H = H_{\text{Fe}} + H_{\text{Tm},0} + \sum_{\nu} H_{K^-, \nu} + H_{B,\text{Fe}} + H_{B,\text{Tm}} + H_{E,\text{Tm}}. \quad (70)$$

Here ν labels inequivalent Fe–Tm geometric bond orbits. Each $H_{K^-, \nu}$ has the form of eq. (60) but with its own reference columns $\mathbf{k}_{\nu,2}$, $\mathbf{k}_{\nu,5}$, $\mathbf{k}_{\nu,7}$ and its own cell offsets.

13 Higher-order symmetry-allowed couplings involving external fields and two Fe spins

The previous sections derived the lowest-order Fe–Tm bilinear

$$\mathbf{S}_i^T A_b \mathbf{J}_j \longrightarrow \mathbf{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \quad (71)$$

The same logic gives the higher-order couplings. The key rule is that one must first write a physical operator with well-defined spatial and time-reversal transformation properties, and only then project the Tm operator into the local SU(3) qutrit space.

We will use two Tm qutrit sectors:

$$\boldsymbol{\lambda}_j^- = (\lambda_j^2, \lambda_j^5, \lambda_j^7)^T, \quad \text{time-reversal odd}, \quad (72)$$

$$\boldsymbol{\lambda}_j^+ = (\lambda_j^1, \lambda_j^3, \lambda_j^4, \lambda_j^6, \lambda_j^8)^T, \quad \text{time-reversal even}. \quad (73)$$

Often only the off-diagonal time-even coherences are kept,

$$\boldsymbol{\lambda}_{j,\text{off}}^+ = (\lambda_j^1, \lambda_j^4, \lambda_j^6)^T. \quad (74)$$

The local mirror m_z acts on the three CEF states as

$$W_{m_z} = \text{diag}(+1, +1, -1). \quad (75)$$

Therefore the mirror/parity matrices in the local qutrit basis are

$$M_- = \text{diag}(+1, -1, -1), \quad \text{basis } (\lambda^2, \lambda^5, \lambda^7), \quad (76)$$

$$M_+ = \text{diag}(+1, +1, -1, -1, +1), \quad \text{basis } (\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8), \quad (77)$$

$$M_{+,\text{off}} = \text{diag}(+1, -1, -1), \quad \text{basis } (\lambda^1, \lambda^4, \lambda^6). \quad (78)$$

Key point

Time reversal fixes which qutrit sector can appear:

Factor	$\mathcal{T}$ parity	Inversion type	Comment
$\mathbf{S}$	odd	axial, even under I	Fe spin
$\mathbf{J}$	odd	axial, even under I	physical Tm angular momentum
$\mathbf{B}$	odd	axial, even under I	magnetic field
$\mathbf{E}$	even	polar, odd under I	electric field
λ^-	odd	M_-	$\lambda^2, \lambda^5, \lambda^7$
λ^+	even	M_+	$\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$

A term in the Hamiltonian must be time-reversal even unless the external nonequilibrium drive is being treated as an explicitly time-dependent source. Thus ESJ uses the λ^- sector after PJP projection, while $BS\lambda$ and $SS\lambda$ naturally use the λ^+ sector.

13.1 Electric-field-assisted Fe–Tm term: $E + S + J$

Start from a physical third-order coupling on a bond $b = (i, j; \mathbf{n})$,

$$H_b^{ESJ}(t) = \sum_{\eta, \alpha, \beta} E_\eta(t) S_{i, \alpha} C_{b; \eta \alpha \beta}^{EJ} J_{j, \beta}. \quad (79)$$

Here η is a polar-vector electric-field index, α is an Fe-spin axial-vector index, and β is a physical Tm angular-momentum axial-vector index. Since E is time-even while S and J are time-odd, the product ESJ is time-even.

Now project the physical J operator:

$$P_j J_\beta P_j = \sum_{a \in \{2, 5, 7\}} \mu_{j, \beta a} \lambda_j^a. \quad (80)$$

Then

$$P_j H_b^{ESJ} P_j = \sum_{\eta, \alpha, a \in \{2, 5, 7\}} E_\eta(t) S_{i, \alpha} K_{b; \eta \alpha a}^{E, -} \lambda_j^a, \quad (81)$$

with

$$K_{b; \eta \alpha a}^{E, -} = \sum_{\beta} C_{b; \eta \alpha \beta}^{EJ} \mu_{j, \beta a}. \quad (82)$$

Thus $E + S + J$ is not a new kind of Tm operator; after projection it is an electric-field-assisted source for the same time-odd qutrit operators $\lambda^2, \lambda^5, \lambda^7$.

Gauge-covariant symmetry rule. For a space-group operation g , the polar electric field transforms with the polar matrix Q_g , while S and J transform with the axial matrix R_g . In the projected qutrit basis,

$$U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{\sigma_g(j)}^b. \quad (83)$$

Therefore the projected tensor satisfies the index rule

$$K_{gb; \nu \gamma c}^{E, -} = \sum_{\eta, \alpha, a} Q_{g, \eta \nu} R_{g, \alpha \gamma} K_{b; \eta \alpha a}^{E, -} \left[D_g^{(j)} \right]_{ca}. \quad (84)$$

This is the field-assisted analogue of the K^- covariance rule. The extra Q_g index is the only new ingredient.

Inversion partner. For inversion, $Q_I = -1$ because E is polar, while $R_I = +1$ for axial vectors. In the transported gauge the relative Tm operation gives M_- , so the inversion partner carries

$$\boxed{\text{odd-bond sign matrix for } ESJ : \quad -M_- = \text{diag}(-1, +1, +1).} \quad (85)$$

Thus the selection rule is the opposite of the static K^- term:

$$\lambda^2 : \text{opposite sign on the inversion partner}, \quad (86)$$

$$\lambda^5, \lambda^7 : \text{same sign on the inversion partner}. \quad (87)$$

Transported-gauge orbit formula. Let the reference electric-assisted column vectors be

$$\mathbf{e}_{\eta,2}, \quad \mathbf{e}_{\eta,5}, \quad \mathbf{e}_{\eta,7}, \quad (88)$$

where each is a three-component Fe-spin vector. For the proper coset sending the reference Fe sublattice to i , define Q_i as the polar point matrix and R_i as the axial spin matrix. A compact transported-gauge form is

$$\boxed{H_{\text{orbit}}^{ESJ}(t) = \sum_{i=0}^3 \sum_{\nu} E_{\nu}(t) \mathbf{S}_i^T R_i \sum_{\eta} Q_{i,\eta\nu} \left[\mathbf{e}_{\eta,2}(\lambda_i^2 - \lambda_{3-i}^2) \right.} \quad (89)$$

$$\left. + \mathbf{e}_{\eta,5}(\lambda_i^5 + \lambda_{3-i}^5) + \mathbf{e}_{\eta,7}(\lambda_i^7 + \lambda_{3-i}^7) \right].$$

If the code has already contracted the electric tensor onto a fixed lab polarization, the factor $\sum_{\eta} Q_{i,\eta\nu} E_{\nu}$ should be viewed as a sublattice-dependent effective coefficient. Dropping it is only valid if the coupling parameters have already been fitted separately for each proper-coset partner.

Implementation prescription

For an implementation storing Fe spins in global axes and Tm operators in local qutrit bases, the ESJ odd-bond signs are

$$\lambda^2 : -1, \quad \lambda^5 : +1, \quad \lambda^7 : +1. \quad (90)$$

Additionally, proper-coset partners require the Fe row signs R_i , and field-assisted terms require the electric-field polar-index signs Q_i unless the field polarization has already been folded into orbit-dependent coefficients.

13.2 Magnetic-field-assisted Fe–Tm term: $B + S + \lambda$

A magnetic-field-assisted coupling with one Fe spin and one Tm qutrit operator has the form

$$\boxed{H_b^{BS\lambda}(t) = \sum_{\eta,\alpha,a \in +} B_{\eta}(t) S_{i,\alpha} K_{b;\eta\alpha}^{B,+} \lambda_j^a.} \quad (91)$$

Here $a \in +$ means

$$a \in \{1, 3, 4, 6, 8\}, \quad (92)$$

or, if only off-diagonal coherences are retained,

$$a \in \{1, 4, 6\}. \quad (93)$$

This time-even sector is required because B and S are both time-reversal odd, so BS is time-reversal even.

If one wants to derive this term from a physical Tm time-even operator $\mathcal{O}_\rho^+$, write

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0} \mathbf{1} + \sum_{a \in \{1,3,4,6,8\}} o_{j,\rho a} \lambda_j^a. \quad (94)$$

A physical coupling

$$\sum_{\eta, \alpha, \rho} B_\eta S_{i,\alpha} L_{b;\eta\alpha\rho} \mathcal{O}_{j,\rho}^+ \quad (95)$$

projects to eq. (91) with

$$K_{b;\eta\alpha a}^{B,+} = \sum_{\rho} L_{b;\eta\alpha\rho} o_{j,\rho a}. \quad (96)$$

This is the even-sector analogue of the PJP step.

Gauge-covariant symmetry rule. The magnetic field is an axial vector, so it transforms with R_g , not Q_g . Therefore

$$K_{gb;\nu\gamma c}^{B,+} = \sum_{\eta, \alpha, a} R_{g,\eta\nu} R_{g,\alpha\gamma} K_{b;\eta\alpha a}^{B,+} \left[D_{g,+}^{(j)} \right]_{ca}. \quad (97)$$

Here $D_{g,+}$ is the adjoint action restricted to the time-even qutrit sector.

Inversion partner. Under inversion, B is axial and hence does not change sign. S is also axial. Thus the odd-bond sign is just the time-even qutrit mirror matrix:

$$\text{odd-bond sign matrix for } BS\lambda^+ : \quad M_+ = \text{diag}(+1, +1, -1, -1, +1). \quad (98)$$

For the off-diagonal subset $(\lambda^1, \lambda^4, \lambda^6)$,

$$M_{+, \text{off}} = \text{diag}(+1, -1, -1). \quad (99)$$

So

$$\lambda^1, \lambda^3, \lambda^8 : \text{same sign on the inversion partner}, \quad (100)$$

$$\lambda^4, \lambda^6 : \text{opposite sign on the inversion partner}. \quad (101)$$

Transported-gauge orbit formula, off-diagonal version. For reference column vectors $\mathbf{b}_{\eta,1}$, $\mathbf{b}_{\eta,4}$ and $\mathbf{b}_{\eta,6}$,

$$H_{\text{orbit,off}}^{BS\lambda}(t) = \sum_{i=0}^3 \sum_{\nu} B_{\nu}(t) \mathbf{S}_i^T R_i \sum_{\eta} R_{i,\eta\nu} \left[\mathbf{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) + \mathbf{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \mathbf{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6) \right]. \quad (102)$$

If the diagonal even operators are included, add

$$\mathbf{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8). \quad (103)$$

to the bracket.

Implementation prescription

For $BS\lambda^+$, the inversion signs are not the same as the static K^- term. They are

$$\lambda^1 : +1, \quad \lambda^4 : -1, \quad \lambda^6 : -1 \quad (104)$$

for the off-diagonal sector. Proper-coset partners require Fe row signs R_i and magnetic-field axial-index signs R_i .

13.3 Fe–Fe–Tm term: $S + S + \lambda$

A symmetry-allowed term with two Fe spins and one Tm qutrit operator is

$$H_b^{SS\lambda} = \sum_{m,n} \sum_{\alpha,\beta,a \in +} S_{m,\alpha} S_{n,\beta} W_{b;mn;\alpha\beta a} \lambda_j^a. \quad (105)$$

Because two Fe spins are time-reversal even as a product, the Tm operator must again be time-reversal even:

$$a \in \{1, 3, 4, 6, 8\}. \quad (106)$$

For the onsite version used in many minimal simulations, $m = n = i$ and

$$H_{b,\text{onsite}}^{SS\lambda} = \sum_{a \in +} \mathbf{S}_i^T W_b^a \mathbf{S}_i \lambda_j^a. \quad (107)$$

For a single classical spin on the same site, only the symmetric part of W^a matters, so one may parameterize

$$W^a = \begin{pmatrix} W_{xx}^a & W_{xy}^a & W_{xz}^a \\ W_{xy}^a & W_{yy}^a & W_{yz}^a \\ W_{xz}^a & W_{yz}^a & W_{zz}^a \end{pmatrix}. \quad (108)$$

This is the origin of the six real parameters (xx, yy, zz, xy, xz, yz) per λ^a channel in a code implementation.

If the term is derived from a physical time-even Tm multipole $\mathcal{O}_\rho^+$, then

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0} \mathbf{1} + \sum_{a \in +} o_{j,\rho a} \lambda_j^a, \quad (109)$$

and the physical coupling

$$\sum_{\alpha,\beta,\rho} S_{m,\alpha} S_{n,\beta} W_{b;mn;\alpha\beta\rho} \mathcal{O}_{j,\rho}^+ \quad (110)$$

projects to eq. (105) with

$$W_{b;mn;\alpha\beta a} = \sum_{\rho} W_{b;mn;\alpha\beta\rho} o_{j,\rho a}. \quad (111)$$

Gauge-covariant symmetry rule. For the general two-Fe-site case,

$$W_{gb;\gamma\delta c} = \sum_{\alpha,\beta,a} R_{g,\alpha\gamma} R_{g,\beta\delta} W_{b;\alpha\beta a} \left[D_{g,+}^{(j)} \right]_{ca}. \quad (112)$$

For the onsite matrix form $\mathbf{S}^T W^a \mathbf{S}$, this becomes

$$W_{gb}^c = \sum_a \left[D_{g,+}^{(j)} \right]_{ca} R_g^T W_b^a R_g. \quad (113)$$

Since the R_i used here are diagonal π -rotation matrices, $R_i^T = R_i$.

Inversion partner. There is no polar electric prefactor and no odd physical J operator. Hence the inversion sign is again just the time-even qutrit parity matrix M_+ :

$$\boxed{\text{odd-bond sign matrix for } SS\lambda^+ : \quad M_+ = \text{diag}(+1, +1, -1, -1, +1).} \quad (114)$$

Thus $\lambda^1, \lambda^3, \lambda^8$ couple to inversion-even Tm combinations, whereas λ^4, λ^6 couple to inversion-odd Tm combinations.

Transported-gauge onsite orbit formula. Keeping only the off-diagonal even coherences for compactness,

$$\boxed{H_{\text{orbit,off}}^{SS\lambda} = \sum_{i=0}^3 \left[\mathbf{S}_i^T R_i W^1 R_i \mathbf{S}_i (\lambda_i^1 + \lambda_{3-i}^1) \right.} \\ \left. + \mathbf{S}_i^T R_i W^4 R_i \mathbf{S}_i (\lambda_i^4 - \lambda_{3-i}^4) \right.} \\ \left. + \mathbf{S}_i^T R_i W^6 R_i \mathbf{S}_i (\lambda_i^6 - \lambda_{3-i}^6) \right].} \quad (115)$$

If λ^3 and λ^8 are retained, add

$$\mathbf{S}_i^T R_i W^3 R_i \mathbf{S}_i (\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{S}_i^T R_i W^8 R_i \mathbf{S}_i (\lambda_i^8 + \lambda_{3-i}^8). \quad (116)$$

13.4 Sign-summary table for implementation

For one transported-gauge inversion pair, the relative sign on the odd bond is the product of three contributions:

$$\boxed{\text{odd-bond sign} = (\text{external-field inversion parity}) \times (\text{local qutrit mirror parity}).} \quad (117)$$

Fe spin is axial, so it does not add an inversion sign. However, proper cosets still require R_i row signs for Fe components if Fe spins are stored in global axes.

Term	Tm sector	Odd-bond sign matrix	Surviving inversion-even Tm combination
$SJ \rightarrow S\lambda^-$	(2, 5, 7)	$M_- = (+, -, -)$	λ^2
$ESJ \rightarrow ES\lambda^-$	(2, 5, 7)	$-M_- = (-, +, +)$	λ^5, λ^7
$BS\lambda_{\text{off}}^+$	(1, 4, 6)	$M_{+, \text{off}} = (+, -, -)$	λ^1
$SS\lambda_{\text{off}}^+$	(1, 4, 6)	$M_{+, \text{off}} = (+, -, -)$	λ^1
$BS\lambda^+ \text{ or } SS\lambda^+$	(1, 3, 4, 6, 8)	$(+, +, -, -, +)$	$\lambda^1, \lambda^3, \lambda^8$

Warning

The phrase “surviving inversion-even Tm combination” refers only to the pair sum/ difference induced by the inversion partner. It does not say that the full response is nonzero for every Fe magnetic pattern or every field polarization. Proper-coset row signs, field-index signs, and the actual Fe mode pattern can impose additional cancellations.

13.5 Complete Hamiltonian with every term written out

Collecting the Fe sector, the onsite Tm sector, the leading bilinear K^- , the three higher-order field/two-spin vertices, and the external-field drives, the full transported-gauge model is

$$\boxed{H(t) = H_{\text{Fe}} + H_{\text{Tm},0} + \sum_{\nu} \left(H_{\nu}^{K^-} + H_{\nu}^{ESJ}(t) + H_{\nu}^{BS\lambda}(t) + H_{\nu}^{SS\lambda} \right) + H_{B,\text{Fe}}(t) + H_{B,\text{Tm}}(t) + H_{E,\text{Tm}}(t),} \quad (118)$$

where ν runs over inequivalent Fe-Tm bond orbits and $i = 0, 1, 2, 3$ over the four sublattices, with the proper-coset row matrices

$$R_0 = \text{diag}(+, +, +), \quad R_1 = \text{diag}(+, -, -), \quad R_2 = \text{diag}(-, +, -), \quad R_3 = \text{diag}(-, -, +). \quad (119)$$

Every term is now written explicitly.

Fe sector (first/second neighbour exchange and single-ion anisotropy).

$$H_{\text{Fe}} = \sum_{\langle mn \rangle_1} [J_{1xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{1z} S_m^z S_n^z] + \sum_{\langle mn \rangle_2} [J_{2xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{2z} S_m^z S_n^z] \\ + \sum_m [D_x (S_m^x)^2 + D_y (S_m^y)^2 + D_z (S_m^z)^2]. \quad (120)$$

Onsite Tm sector (three-singlet qutrit).

$$H_{\text{Tm},0} = \sum_j \left[-\frac{\Delta_{12}}{2} \lambda_j^3 + \frac{\Delta_{12}-2\Delta_{13}}{2\sqrt{3}} \lambda_j^8 \right], \quad \Delta_{12} \approx 0.52 \text{ THz}, \quad \Delta_{13} \approx 1.13 \text{ THz}. \quad (121)$$

Static bilinear K^- (columns $\mathbf{k}_{\nu,2}, \mathbf{k}_{\nu,5}, \mathbf{k}_{\nu,7}$), λ^- sector.

$$H_{\nu}^{K^-} = \sum_{i=0}^3 \mathbf{S}_i^T R_i [\mathbf{k}_{\nu,2}(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_{\nu,5}(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_{\nu,7}(\lambda_i^7 - \lambda_{3-i}^7)]. \quad (122)$$

Electric-field-assisted ESJ (columns $\mathbf{e}_{\eta,a}$), λ^- sector.

$$H_{\nu}^{ESJ}(t) = \sum_{i=0}^3 \sum_{\eta} E_{\eta}(t) \mathbf{S}_i^T R_i [\mathbf{e}_{\eta,2}(\lambda_i^2 - \lambda_{3-i}^2) + \mathbf{e}_{\eta,5}(\lambda_i^5 + \lambda_{3-i}^5) + \mathbf{e}_{\eta,7}(\lambda_i^7 + \lambda_{3-i}^7)], \quad (123)$$

where the polar field carries $\mathbf{e}_{\eta,a} \rightarrow \sum_{\kappa} Q_{i,\eta\kappa} \mathbf{e}_{\kappa,a}$ with $Q_I = -1$ (this is the origin of the relative $-M_-$ versus K^-).

Magnetic-field-assisted $BS\lambda$ (columns $\mathbf{b}_{\eta,a}$), λ^+ sector.

$$H_{\nu}^{BS\lambda}(t) = \sum_{i=0}^3 \sum_{\eta} B_{\eta}(t) \mathbf{S}_i^T R_i \left[\mathbf{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) + \mathbf{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \mathbf{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6) \right. \\ \left. + \mathbf{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8) \right], \quad (124)$$

with $\mathbf{b}_{\eta,a} \rightarrow \sum_{\kappa} R_{i,\eta\kappa} \mathbf{b}_{\kappa,a}$ (axial field, hence R_i not Q_i).

Two-Fe-spin $SS\lambda$ (onsite symmetric matrices W^a), λ^+ sector.

$$H_{\nu}^{SS\lambda} = \sum_{i=0}^3 \left[(\mathbf{S}_i^T R_i W^1 R_i \mathbf{S}_i)(\lambda_i^1 + \lambda_{3-i}^1) + (\mathbf{S}_i^T R_i W^4 R_i \mathbf{S}_i)(\lambda_i^4 - \lambda_{3-i}^4) + (\mathbf{S}_i^T R_i W^6 R_i \mathbf{S}_i)(\lambda_i^6 - \lambda_{3-i}^6) \right. \\ \left. + (\mathbf{S}_i^T R_i W^3 R_i \mathbf{S}_i)(\lambda_i^3 + \lambda_{3-i}^3) + (\mathbf{S}_i^T R_i W^8 R_i \mathbf{S}_i)(\lambda_i^8 + \lambda_{3-i}^8) \right]. \quad (125)$$

External-field drives.

$$H_{B,\text{Fe}}(t) = g_{\text{Fe}}\mu_B \sum_m \mathbf{B}(t) \cdot \mathbf{S}_m, \quad (126)$$

$$H_{B,\text{Tm}}(t) = -g_J\mu_B \sum_j \sum_{a \in \{2,5,7\}} \left[\sum_{\alpha,\beta} B_\alpha(t) (R_j)_{\alpha\beta} \mu_{0,\beta a} \right] \lambda_j^a, \quad (127)$$

$$H_{E,\text{Tm}}(t) = \sum_j \sum_{a \in \{1,3,4,6,8\}} \left[\sum_\eta E_\eta(t) p_{j,\eta a} \right] \lambda_j^a, \quad (128)$$

where μ_0 is the projected-moment matrix (19) and $p_{j,\eta a}$ is the projected Tm electric-multipole tensor (time-even, hence λ^+).

Time-reversal sector bookkeeping. The λ^- sector $\{2, 5, 7\}$ is fed by K^- and ESJ ; the λ^+ sector $\{1, 3, 4, 6, 8\}$ is fed by $BS\lambda$, $SS\lambda$, and $H_{E,\text{Tm}}$. $H_{\text{Fe}}, H_{\text{Tm},0}, H^{K^-}, H^{SS\lambda}, H_{B,\text{Fe}}, H_{B,\text{Tm}}$ are time-even by construction; the field-assisted terms are time-even only together with their explicit external field. This sorting is exactly what the cross-peak analysis of section 16 acts on.

14 Implementation remedy

14.1 If Fe spins are stored in global axes

Use the transported-gauge local λ_j basis for Tm. Then for each bond in a symmetry orbit:

$$\boxed{K_{\text{bond}}^- = R_{\text{Fe}(\text{bond})} K_\nu^- M_{\text{inv}},} \quad (129)$$

where

$$M_{\text{inv}} = \begin{cases} \mathbf{1}, & \text{direct/proper-coset branch,} \\ M_- = \text{diag}(+1, -1, -1), & \text{inversion-related branch.} \end{cases} \quad (130)$$

This is the version that matches a code convention in which Fe spins are lab Cartesian and Tm SU(3) states are local CEF-basis variables.

14.2 If Fe spins are instead stored in sublattice local frames

Then the R_i row signs may be absorbed into the spin variables themselves. In that convention the bond tensor should not multiply by R_i again. The code and the theory must choose one convention and use it consistently.

Warning

Do not mix these conventions. If Fe spins are global, include the R_i row signs in the bond tensor. If Fe spins are local, do not include them again. The Tm λ_j operators remain local qutrit coordinates either way.

14.3 Do not use $\mu^{-1}R\mu$ as a qutrit frame by default

The safe object is $P_j J_\alpha P_j$ or, in local coordinates, $\mu_{j,\alpha a} \lambda_j^a$. If a global observable is needed, compute it through

$$J_{\alpha,j}^{\text{global}} = \sum_{\beta,a} (R_j)_{\alpha\beta} \mu_{0,\beta a} \lambda_j^a \quad (131)$$

in transported gauge, or through the fully general W_g, D_g construction. Avoid treating

$$\mu^{-1}R\mu \quad (132)$$

as a genuine SU(3) adjoint representation unless it has been verified to be a proper orthogonal adjoint action induced by a unitary W .

15 Diagrammatic summary

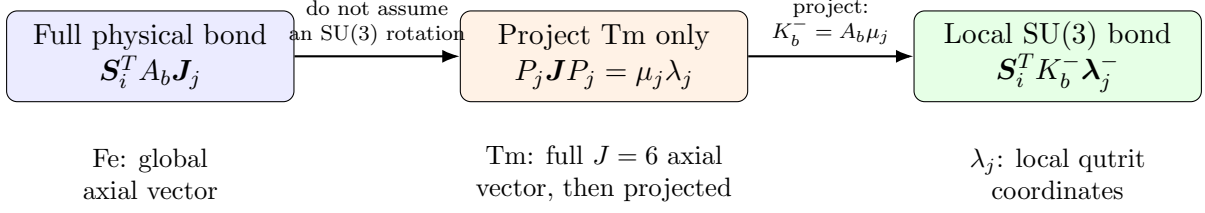


Figure 1: The correct order of operations. The Tm angular momentum is projected only after the physical symmetry constraints are formulated.

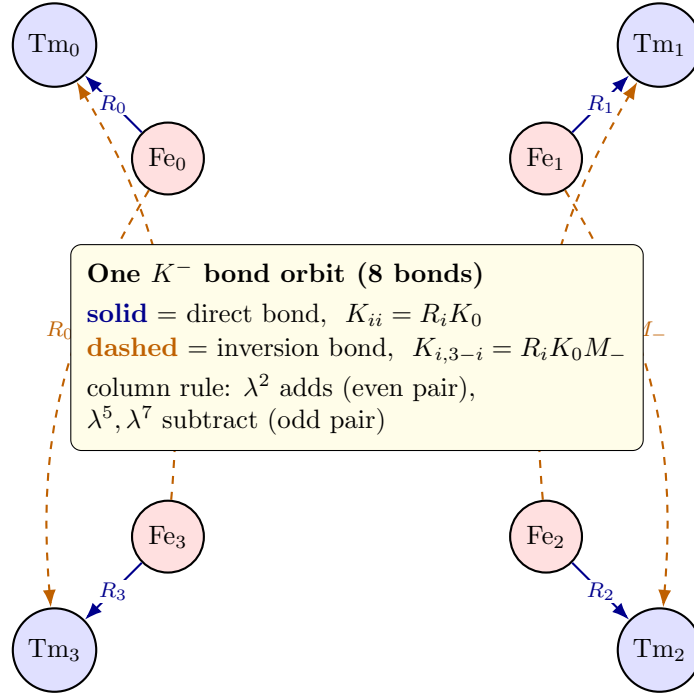


Figure 2: One transported-gauge K^- orbit. Fe sites (red) sit on inversion centres; Tm sites (blue) are exchanged in pairs by inversion ($\text{Tm}_0 \leftrightarrow \text{Tm}_3$, $\text{Tm}_1 \leftrightarrow \text{Tm}_2$). Each Fe participates in one *direct* bond (solid, coefficient $R_i K_0$) to its own Tm and one *inversion* bond (dashed, coefficient $R_i K_0 M_-$) to the partner Tm. Summing the orbit gives the even/odd column rule of (60).

16 Modeling the 2DCS cross peak: what must a vertex do?

The earlier sections fixed the *form* of every symmetry-allowed Fe–Tm coupling. We now ask the sharp dynamical question: which of them can actually produce the measured (q_{AFM}, E_{12}) 2DCS cross peak? The strategy is two-step. First we reduce the whole experiment to a *single* sharp requirement on an effective Tm force (this section). Then we test each candidate coupling against it as a Feynman-diagram catalogue (section 17). The experimental context is third-order magnetic-dipole THz 2DCS of orthoferrite magnons [1]; the language is standard nonlinear response theory [2].

16.1 The observable and M_{NL} , precisely

Two collinear THz pulses give the total drive $B(t) = B_A(t+\tau) + B_B(t)$. The detected magnetization is a functional of that field; expand it in powers of the drive,

$$M[B] = \sum_{n \geq 1} M^{(n)}[\underbrace{B, \dots, B}_n]. \quad (133)$$

With $M_A \equiv M[B_A]$, $M_B \equiv M[B_B]$, $M_{AB} \equiv M[B_A + B_B]$, the isolated nonlinear signal is

$$M_{\text{NL}}(t, \tau) = M_{AB} - M_A - M_B. \quad (134)$$

Order by order, $M^{(n)}[B_A + B_B] - M^{(n)}[B_A] - M^{(n)}[B_B]$ keeps only the *mixed* terms, those with at least one factor of each pulse:

$$n = 1 : \quad 0 \quad (\text{linear, no mixed term}), \quad (135)$$

$$n = 2 : \quad 2 M^{(2)}[B_A, B_B] \quad (\chi^{(2)}\text{-type, } \propto B_A B_B), \quad (136)$$

$$n = 3 : \quad M^{(3)} \text{ mixed} \quad (\chi^{(3)}\text{-type, } B_A^2 B_B + B_A B_B^2). \quad (137)$$

So M_{NL} is purely nonlinear and starts at second order; the $n = 1$ piece cancels identically. For magnetic-dipole driving in a centrosymmetric magnet the leading robust contribution is third order [1]. The detected quantity at the target is the Tm c -axis dipole, $m_z^{\text{Tm}} \propto \lambda^2$, so the object of interest is $M_{\text{NL}}^{\lambda^2}(t, \tau)$, whose 2D transform should peak at $(\omega_\tau, \omega_t) = (q_{\text{AFM}}, E_{12})$.

16.2 The detection channel: only λ^2 is bright

In the E_{12} block the coherence is carried by $\lambda^1 = \text{Re}|1\rangle\langle 2|$ and $\lambda^2 = \text{Im}|1\rangle\langle 2|$, both oscillating at $\omega_{12} = E_{12}$, while λ^3 is the population imbalance. From (19) the radiated moment is

$$m_z^{\text{Tm}} \propto P_j J_z P_j = 5.264 \lambda^2, \quad (138)$$

so *only λ^2 is magnetically bright*. The onsite Hamiltonian $H_{\text{Tm},0} \supset -\frac{\Delta_{12}}{2} \lambda^3$ generates, through $[\lambda^3, \lambda^1] = 2i\lambda^2$, the planar precession

$$\dot{\lambda}^1 = -\Delta_{12} \lambda^2, \quad \dot{\lambda}^2 = +\Delta_{12} \lambda^1. \quad (139)$$

A force on *either* λ^1 or λ^2 therefore launches the same ring at ω_{12} . We may treat the Tm E_{12} sector as a single driven oscillator,

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t-t')] h^1(t') dt', \quad (140)$$

driven by the effective transverse field $h^1(t)$ that a conversion vertex exerts on λ^1 .

16.3 Reframing to a sharp question

Equation (140) factorizes the 2D map. The *detection*-axis (t) structure is fixed entirely by the Tm pole at E_{12} : an impulsive step force gives a $1 - \cos \omega_{12}t$ ring, a delta kick gives a $\sin \omega_{12}t$ ring. All the *delay*-axis (τ) information—in particular whether the peak sits at $\omega_\tau = q_{\text{AFM}}$ —lives in the τ -dependence of $h^1(t, \tau)$. Hence the entire experiment collapses to one question:

The sharp question

A coupling produces the (q_{AFM}, E_{12}) cross peak *iff* the effective Tm force it generates,

$$M_{\text{NL}}^{\lambda^2}(\tau, t) \propto f(\omega_q \tau) [1 - \cos \omega_{12}t], \quad (141)$$

has f oscillating at $\omega_q = q_{\text{AFM}}$. Equivalently, the force $h^1(t, \tau)$ on the bright λ^1 channel must be (i) nonzero by symmetry, (ii) able to reach λ^1/λ^2 , and (iii) carry an ω_q modulation along the delay τ .

Everything below is the evaluation of $h^1(t, \tau)$ for each candidate vertex.

16.4 Feynman rules: fields, charges, and the vertex requirement

We treat the problem as a tree-level field theory. The propagating “fields” (internal/external lines) are

line	object	properties
Fe magnon	δS_α	pole at ω_q ; time-odd; the pump excites δS_y
Tm qutrit	λ^a	poles at CEF gaps; $\lambda^- = \{2, 5, 7\}$ (T-odd), $\lambda^+ = \{1, 3, 4, 6, 8\}$ (T-even)
B-photon	$B_\eta(t)$	external; time-odd, axial
E-photon	$E_\eta(t)$	external; time-even, polar

Every Fe magnon line is born from a pulse through the Zeeman source $g_{\text{Fe}}\mu_B \mathbf{B} \cdot \mathbf{S}$, so each external Fe leg costs one factor of the drive. Three “charges” are conserved at every vertex:

charge	rule at a vertex
energy ω	a static vertex conserves it $\Rightarrow$ connects only equal frequencies; an external leg injects $\hbar\omega_{\text{drive}}$
$\mathcal{T}$ -parity	each term is time-even: λ^- needs an <i>odd</i> number of T-odd legs (S, B); λ^+ an <i>even</i> number
irrep Γ	$\Gamma(\text{legs}) \otimes \Gamma(\lambda) \supset \Gamma_1$; the orbit sign sums of (60) are this projector

Detection and the M_{NL} subtraction add two more bookkeeping rules: the vertex must reach λ^1/λ^2 , and the diagram must attach at least one leg to each pulse. Collecting, a successful cross-peak vertex must pass all six tests:

Scorecard R1–R6

R1 (detection) reaches λ^1/λ^2 (the E_{12} block).

R2 (drive) is sourced by the pumped δS_y magnon.

R3 (parity) is $\mathcal{T}$ -even, which fixes the $\lambda^\pm$ sector.

R4 (irrep) is Γ_1 -allowed for the Γ_2/q_{AFM} geometry.

R5 (energy) can carry the $0.9 \rightarrow 0.5$ THz mismatch—needs ≥ 2 non- λ legs (two magnons, or one magnon + one photon).

R6 (M_{NL}) involves both pulses.

17 The vertex catalogue: a Feynman diagram for each coupling

We now write each candidate as an interaction vertex—a term in H with a coupling constant and a fixed set of legs—draw its tree diagram for the 2DCS process, and score it against R1–R6. In every diagram time runs left to right: a red solid line is an Fe magnon (ω_q), a blue dashed line a Tm coherence, a wavy arrow an external-field interaction (one pulse); a filled dot is a field-gated bilinear vertex, a filled square a two-spin vertex. The boxed expression above each diagram is the vertex itself.

17.1 K^- : the static bilinear (fails R3/R4/R5)

$$H_{K^-} = K_{a\alpha}^- S_\alpha \lambda^a, \quad a \in \{2, 5, 7\} \quad [1 \text{ Fe (odd)} + 1 \text{ Tm}^-(\text{odd}); \mathcal{T}\text{-even; no spare leg}] \quad (142)$$

This is a two-point Fe–Tm vertex; the only process it generates is a single Fe→Tm line conversion (fig. 3). It fails in three independent ways:

1. **(R5, linear).** K^- is linear in S , so it adds to the *quadratic* Hamiltonian—it is a basis rotation among the (harmonic) Fe and Tm normal modes. Coupled harmonic oscillators are diagonalizable and their 2D spectrum has *diagonal peaks only*. A single bilinear hop cannot transport a 0.9 THz coherence into a 0.5 THz one: a static vertex conserves energy, so it can only connect degenerate modes, i.e. it *hybridizes* rather than *converts*. The lone-pulse hop is in any case removed by the M_{NL} subtraction (R6).
2. **(R4, Γ_1).** Even the static torque vanishes: for the patterns F_x, G_z and the pumped δS_y the four-site orbit sums give $[0, 0, 0, 0]$ for every channel except $K_{2y}^- \rightarrow [8, 8, 8, 8]$ (section 18). All channels but $2y$ are Γ_1 -forbidden.
3. **(reorientation).** The one Γ_1 -allowed channel, K_{2y}^- , is a transverse molecular field on the order parameter; it reorients $\Gamma_2 \rightarrow G_y$ before any spectroscopic regime (lost between 0.085 and 0.090 meV). It is not a perturbative vertex.

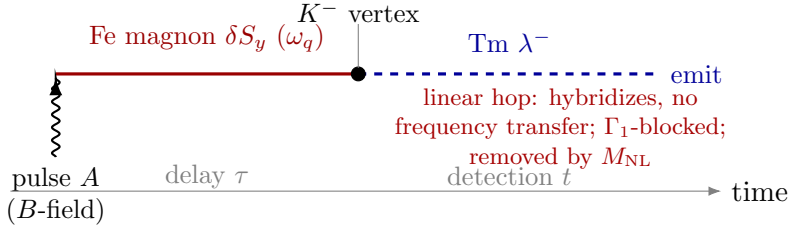


Figure 3: K^- (fails). A single Fe leg converts to a single λ^- leg—a linear hop. It only hybridizes the modes, the conversion matrix element is Γ_1 -cancelled, and the lone-pulse path is subtracted in M_{NL} .

The longitudinal temptation, and why it is still trapped

One might smuggle in two-magnon content through the *longitudinal* component $S_z = S_z^{(0)} - \delta S_\perp^2 / 2S_z^{(0)}$: a bilinear $K S_z \lambda$ does contain $\delta S_y^2 \lambda$. But S_z is time-odd, so $K S_z \lambda$ must place λ in λ^- —it is $K_{2z, 5z, 7z}^-$. The hidden δS_z carries the G_z sublattice sign pattern, so it cancels in the *same* orbit sum as the static G_z torque. The two-magnon signal is real but parity-trapped in λ^- and Γ_1 -cancelled—confirmed by the full nonlinear K_{2z}^- run (which contains the spin-length constraint) being dark to 10^{-15} even at 2.0 meV (section 18). To reach λ^1 the *same* δS_y^2 content must be carried by a $\mathcal{T}$ -even Fe operator: two spin legs (W) or a field leg (κ^B).

17.2 ESJ : electric-field-assisted (works, but wrong target—fails R1)

$$H_{ESJ}(t) = \kappa_{a\alpha\eta}^E E_\eta(t) S_\alpha \lambda^a, \quad a \in \{2, 5, 7\} \quad [1 \text{ E (even)} + 1 \text{ Fe (odd)} + 1 \text{ Tm}^-(\text{odd}); \mathcal{T}\text{-even}] \quad (143)$$

A genuine three-point vertex (fig. 4): pulse A’s magnetic field makes the Fe coherence, pulse B’s *electric* field enters at the vertex, and the E -photon supplies the energy mismatch (R5 ✓, R6 ✓, R2 ✓). But it feeds λ^- , and the surviving inversion-even combination under $-M_-$ is λ^5, λ^7 —the E_{13}

and E_{23} coherences, *not* E_{12} . So ESJ produces real cross peaks at (q_{AFM}, E_{13}) and (q_{AFM}, E_{23}) , but lands on the wrong readout for E_{12} (fails R1).

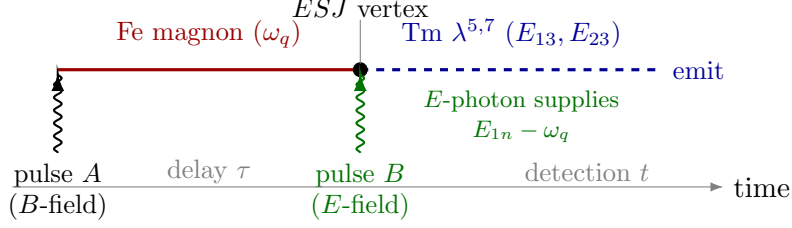


Figure 4: ESJ (works, wrong target). A clean three-point vertex; the E -photon carries the energy mismatch. But it feeds $\lambda^{5,7}$, so it lights up E_{13}/E_{23} , not the E_{12} target.

17.3 $W = SS\lambda$: the two-magnon vertex (passes all—the winner)

$$\boxed{H_W = W_{\alpha\beta}^a S_\alpha S_\beta \lambda^a, \quad a \in \{1, 3, 4, 6, 8\}} \quad [2 \text{ Fe (odd} \times \text{odd} = \text{even)} + 1 \text{ Tm}^+ (\text{even}); \mathcal{T}\text{-even; no photon}] \quad (144)$$

For the pump geometry the active term is $W_1^{yy} S_y^2 \lambda^1$ (R2: two y -magnon legs; R1: λ^1). This is a three-point vertex with *two* Fe legs, one sourced by each pulse (R6); see fig. 5. The vertex feels $h^1(t) = W_1^{yy} S_y(t)^2$ with $S_y(t) = A[\sin \omega_q(t + \tau) + \theta(t) \sin \omega_q t]$. The only term surviving M_{NL} is the two-pulse cross product $2S_y^a S_y^b$, which *rectifies*:

$$h_{\text{NL}}^1(t, \tau) = W_1^{yy} A^2 \left[\underbrace{\cos \omega_q \tau}_{\text{difference}} - \underbrace{\cos \omega_q (2t + \tau)}_{\text{sum}} \right] \theta(t). \quad (145)$$

The difference term is a t -independent step, born at the probe, which through (140) launches the $1 - \cos \omega_{12} t$ ring. Hence

$$\boxed{M_{\text{NL}}^W(\tau, t) \propto W_1^{yy} A^2 \cos(\omega_q \tau) [1 - \cos \omega_{12} t].} \quad (146)$$

This is exactly the sharp-question form: it factorizes, oscillating at ω_q along τ and at ω_{12} along t (R3 with the delay modulation at ω_q), giving a single sharp peak at (q_{AFM}, E_{12}) with no diagonal partner (the $2\omega_q$ “sum” companion is off-resonant). The simulated λ^2 map has its rank-1 maximum exactly here, amplitude 51.0, SNR 117, the (E_{12}, E_{12}) diagonal $\sim 50\times$ weaker [7]; yy beats xx by $\sim 600\times$ (R2: S_x is the static F_x , barely fluctuating).

Where the E_{12} energy comes from (verified). Since δS_y^2 contains only 0 and $2\omega_q$, a static three-leg vertex between *free* modes could never emit at ω_{12} . The resolution is that the magnon legs are pulsed envelopes, not free modes: the rectified cross term is a *suddenly switched-on step*, and a step carries spectral weight at all frequencies in its edge, $\propto e^{-\omega_{12}^2 \sigma^2 / 2}$ for rise time σ . Quantum mechanically this is an intra-pulse difference-frequency (impulsive-Raman) process: the E_{12} quantum is drawn from the *pulse bandwidth*, while the magnon pair carries only the phase memory $\cos \omega_q \tau$. The mechanism is displacive and therefore requires nonadiabaticity—a falsifiable statement verified directly (section 19.5): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades ($51.7 \rightarrow 1.7 \times 10^{-3}$), the adiabatic suppression the step picture demands, while switching the direct Tm drive off changes the peak by $< 10^{-4}$ relative (pure- W origin). The same sweep resolves the second Fe leg: for impulsive pulses it is the resonant probe magnon (peak $\propto A^2$); for longer pulses it crosses over to the *field-following quasistatic tilt* $\delta S^{\text{qs}}(t) \propto B(t)$ (peak $\propto A^1$)—both W -mediated, same peak location. The quasistatic leg also enables the *reverse* conversion $\lambda^{1,2}(\tau) \times \delta S^{\text{qs}}(\text{pulse } B) \rightarrow q_{\text{AFM}}$, an (E_{12}, q_{AFM}) feature in the Fe channel whenever a first-order E_{12} delay coherence exists (section 19).

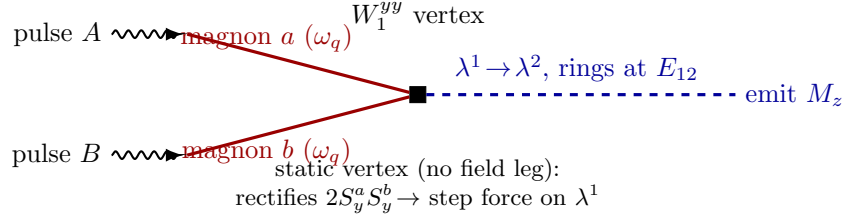


Figure 5: W_1^{yy} (winner). Each pulse makes one δS_y magnon; the two converge on the time-independent $S_y^2 \lambda^1$ vertex (square), whose two-pulse cross term rectifies to a step force that launches the $\lambda^1 \rightarrow \lambda^2$ ring at E_{12} . The response factorizes, $\cos(\omega_q \tau)[1 - \cos \omega_{12} t]$, giving the clean rank-1 cross peak.

17.4 $\kappa^B = BS\lambda$: magnetic-field-assisted (passes, but noisy)

$$H_{\kappa B}(t) = \kappa_{a\alpha\eta}^B B_\eta(t) S_\alpha \lambda^a, \quad a \in \{1, 3, 4, 6, 8\} \quad [1 \text{ B (odd)} + 1 \text{ Fe (odd)} + 1 \text{ Tm}^+(\text{even}); \mathcal{T}\text{-even}] \quad (147)$$

The active term is $\kappa_{1y}^B B_y S_y \lambda^1$. A three-point vertex with one B -photon, one Fe leg and one λ^+ leg (fig. 6); the B -photon supplies the mismatch (R5). It is field-gated: $h^1(t) = \kappa_{1y}^B \mathcal{B}_y(t) S_y(t)$ with $\mathcal{B}_y(t)$ sharply localized at the pulses. At the probe $t = 0$, $S_y(0) = A \sin \omega_q \tau$, so the kick $\propto \sin \omega_q \tau$ launches the same ring,

$$M_{\text{NL}}^{\kappa B}(\tau, t) \propto \kappa_{1y}^B \sin(\omega_q \tau) [1 - \cos \omega_{12} t], \quad (148)$$

again on target (R1–R6 ✓). This is the $BS\lambda^+$ analogue of ESJ , feeding λ^1 instead of $\lambda^{5,7}$. The price: the same gated vertex also dumps each pulse’s *local* Fe response into Tm at both pulse times, generating a strong (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background that dominate the bare cross peak (the on-target maximum is genuine but only rank-3 at zero carrier). A carrier resonant with a Tm transition cleans this up—a sweep amplifies the target $\sim 13\times$ at $\Omega_B = 0.70 \text{ THz}$ (E_{23}) [7]. The definitive resolution, however, is not a carrier but the *polarization* of the gate: the physical vertex is gated by the lab-frame field component $B_\eta(t)$, and its dominant symmetry slice is z -gated, so in the $H \parallel a$ geometry ($B_z = 0$) κ^B is identically silent and its noise never appears. Its proper home is the $H \parallel c$ geometry, where it produces the measured (E_{12}, q_{AFM}) cross peak; see section 19 and fig. 8.

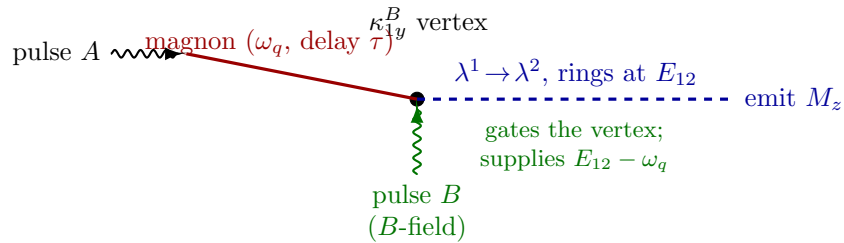


Figure 6: κ_{1y}^B (works, noisy). One magnon from pulse A meets the magnetic field of pulse B at the $S_y \lambda^1$ vertex; the B -photon gates the conversion and carries the mismatch. Same $\lambda^1 \rightarrow \lambda^2$ readout as W_1^{yy} , but it also feeds the (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background, so it needs a resonant carrier to be a robust target.

17.5 Scorecard

vertex	term	legs	sector	R1	R2	R3/4/5	verdict
K^-	$S\lambda^-$	$1\text{Fe}+1\text{Tm}^-$	λ^-	$\times$	—	$\times$	hybridizes / Γ_1 -blocked / reorients
ESJ	$ES\lambda^-$	$E+\text{Fe}+\text{Tm}^-$	λ^-	$\times$	$\checkmark$	$\checkmark$	real, but on E_{13}/E_{23}
W	$SS\lambda^+$	$2\text{Fe}+\text{Tm}^+$	λ^+	$\checkmark$	$\checkmark(yy)$	$\checkmark$	winner: rank-1, factorized
κ^B	$BS\lambda^+$	$B+\text{Fe}+\text{Tm}^+$	λ^+	$\checkmark$	$\checkmark(1y)$	$\checkmark$	works; B_z -gated, geometry 2

The two passes share the same readout, $\lambda^1 \rightarrow \lambda^2$ at E_{12} , and differ only in how the Fe motion is fed in: W uses *two* magnon legs (difference-frequency rectification), κ^B uses *one* magnon leg plus a photon. Both are the minimal $\mathcal{T}$ -even ways to deliver the δS_y^2 two-magnon signal onto the bright λ^1 —the step that K^- , trapped in λ^- , cannot take.

18 Numerical confirmation

The catalogue is confirmed by direct simulation in the calibrated $1 \times 1 \times 1$ mixed TmFeO_3 model. All 2DCS runs use Fe pump $H \parallel x$, $\tau \in [-40, 0]$, $t \in [-40, 40]$, and step $\Delta t = 0.02$ (code units); the Tm sector is not directly pumped unless stated.

18.1 Static one-at-a-time K^- scan (0.2 meV): the Γ_1 block

A static ground-state scan with one K^- component on at a time confirms R4 directly:

component	E/N (meV)	Fe order	$(\lambda_2, \lambda_5, \lambda_7)_{\text{uniform}}$	verdict
baseline	-42.496758	$F_x + G_z$	(0, 0, 0)	reference
K_{2x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{2y}^-	-43.654818	G_y	(0.888320, 0, 0)	spin reorients
K_{2z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5y}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7y}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert

Only K_{2y}^- does anything, and it does so by leaving Γ_2 : an adiabatic ramp keeps Γ_2 stable through 0.085 meV and loses it by 0.090 meV, flipping to G_y with a large uniform λ_2 . The pattern is the exact orbit sign algebra (R_i with $M_- = \text{diag}(+1, -1, -1)$):

$$F_x : K_{2x}^-, K_{5x}^-, K_{7x}^- \rightarrow [0, 0, 0, 0], \quad G_z : K_{2z}^-, K_{5z}^-, K_{7z}^- \rightarrow [0, 0, 0, 0], \quad (149)$$

$$\delta S_y : K_{2y}^- \rightarrow [8, 8, 8, 8], \quad \delta S_y : K_{5y}^-, K_{7y}^- \rightarrow [0, 0, 0, 0]. \quad (150)$$

18.2 Fe-pump 2DCS on the inert channels: dark to machine precision

Running 2DCS on the inert channels (Fe pump only) and extracting the maximum nonlinear $\text{SU}(3)$ amplitude over the (t, τ) grid confirms R5—these are dark even with the *full* nonlinear dynamics (hence including the longitudinal δS_z rectification):

run	$\max M_{\text{NL}}^{\lambda_2} $	$\max M_{\text{NL}}^{\lambda_5} $	$\max M_{\text{NL}}^{\lambda_7} $
$K_{2x}^- = 0.2$	6.06×10^{-18}	0	0
$K_{2x}^- = 1.5$	2.96×10^{-16}	0	0
$K_{2z}^- = 0.2$	9.87×10^{-16}	0	0
$K_{2z}^- = 2.0$	6.86×10^{-15}	0	0
$K_{5y}^- = 0.2$	0	0	0
$K_{7y}^- = 0.2$	0	0	0

Up to $K_{2x}^- = 1.5 \text{ meV}$ and $K_{2z}^- = 2.0 \text{ meV}$ the induced λ_2 is at machine precision—the bilinear cannot, even nonlinearly, manufacture the E_{12} coherence. Adding a direct Tm $H \parallel x$ pump on top reproduces the ordinary Tm response and reveals no hidden K^- channel. By contrast, the no-damping scan over the higher-order set isolates exactly W_1^{yy} (rank-1, $|M_{\text{NL}}^{\lambda_2}| = 51.0$, SNR 117) and κ_{1y}^B as the on-target vertices [7], precisely the two that pass R1–R6.

19 The full scenario: one Hamiltonian, all measured geometries

Section 17 answered the original narrow question—which vertex makes the (q_{AFM}, E_{12}) peak—in a single geometry. The experiment, however, provides *three* pulse/detection geometries, and the full data set is a peak inventory. This section assembles the validated end-to-end picture: one fixed Hamiltonian, physical pulses, and a polarization or selection gate attached to every coupling, such that each geometry self-selects its own subset of cross peaks with no per-geometry retuning. Frequencies are quoted in THz,

$$\omega_F = 0.38 (q_{\text{FM}}), \quad \omega_q = 0.90 (q_{\text{AFM}}), \quad \omega_{12} = 0.50, \quad \omega_{23} = 0.70, \quad \omega_{13} = 1.20 = \omega_{12} + \omega_{23}, \quad (151)$$

the last identity being the shared-level ladder $1 \rightarrow 2 \rightarrow 3$ of the Tm qutrit.

19.1 Geometries, pulse composition, and detection channels

A THz pulse propagating along b carries either $(E \parallel c, B \parallel a)$ or $(E \parallel a, B \parallel c)$. Each field component couples to a fixed set of modes, and each Tm block emits into a fixed polarization channel through the magnetic dipole matrix μ of (19) ($\lambda^2 \rightarrow m_z$ via $\mu_{2z} = 5.264$; $\lambda^5, \lambda^7 \rightarrow m_x$ via $\mu_{5x} = 2.39$, $\mu_{7x} = 0.91$):

	pulse	drives (delay-axis coherences)	detection channels
$\mathcal{G}_A$: $H \parallel a, E \parallel c$	$B \parallel a$	Fe q_{AFM} ($\theta=0$); Tm λ^5, λ^7	<i>cross-pol</i> ($E \parallel a \leftarrow m_z \leftarrow \lambda^2$): geom. 1
	$E \parallel c$	Tm λ^2 (E_{12} c -axis dipole)	<i>same-pol</i> ($E \parallel c \leftarrow m_x \leftarrow \lambda^{5,7}$): geom. 3
$\mathcal{G}_B$: $H \parallel c, E \parallel a$	$B \parallel c$	Fe q_{FM} ($\theta=90^\circ$); Tm λ^2 (μ_{2z})	Fe magnon channel (F_x): geom. 2

Geometries 1 and 3 are therefore the *same physical run* read in the two orthogonal analyzer channels; $B \parallel a$ never reaches λ^2 ($\mu_{2x} = 0$), and $B \parallel c$ never reaches $\lambda^{5,7}$ ($\mu_{5z} = \mu_{7z} = 0$). These two zeros of the μ matrix are the first two gates of the story.

19.2 The mode inventory: the physics of every frequency

Six frequencies carry the entire 2DCS phenomenology. For each we state what oscillates, what sets the frequency, how it is driven, how it radiates, and its linewidth—these five properties, combined with the vertices of section 19.3, determine every peak in every geometry.

q_{FM} ($\omega_F = 0.38$ THz): **the quasi-ferromagnetic magnon.** Precession of the weak net moment $\mathbf{F} \parallel a$ about its equilibrium; in the simulation it is *uniform* δS_y with a staggered δS_x counterpart. The restoring force is the in-plane anisotropy (gap $\propto \sqrt{JK}$), making it the softer magnon. Zeeman-driven only when the field can torque $\mathbf{F}$: $\mathbf{B} \perp \mathbf{M}$, i.e. $H \parallel c$ (the θ selection rule)—which is why it exists only in geometry 2. In $\mathcal{G}_A$ its emission polarization (m_y) is longitudinal to the propagation, and its nonlinear channel is dark to 10^{-11} . 2DCS role: the *delay* coherence of (ω_F, ω_q) in geometry 2.

q_{AFM} ($\omega_q = 0.90$ THz): **the quasi-antiferromagnetic magnon.** Oscillation of the Néel vector: *staggered* $\delta S_y, \delta S_z$ with only a small *uniform* F_x projection inherited from the DM canting. (Practical corollary, learned the hard way: its amplitude must be measured on the staggered magnetization; the uniform $\mathbf{M}$ is blind to it at linear order.) Gap set by exchange $\times$ anisotropy plus DM. Zeeman-driven when $\mathbf{B} \parallel \mathbf{M}$ ($H \parallel a$). 2DCS roles: delay coherence of (ω_q, E_{12}) [G1] and (ω_q, E_{23}) [G3]; *detection* frequency of both geometry-2 cross peaks (read out through the uniform F_x projection); the Fe-channel diagonal.

ω_s (≈ 0.1 THz): **the reorientation soft mode.** Rigid rotation of $\mathbf{F}$ in the a - c plane (uniform S_z)—the $\Gamma_2 \rightarrow \Gamma_4$ reorientation coordinate. Soft because $|K_a - K_c| = 0.003$ meV: the spin-reorientation proximity of TmFeO₃ expressed at $T = 0$. It is neither a delay nor a detection frequency of any listed peak; it appears solely as $\pm\omega_s$ amplitude-modulation sidebands on the strong peaks, carried between sectors by W . Its frequency is a thermometer of the reorientation, $\omega_s(T) \rightarrow 0$ at T_{SR} .

E_{12} ($\omega_{12} = 0.50$ THz): **the ground-doublet coherence.** The $|1\rangle \leftrightarrow |2\rangle$ coherence of the Tm qutrit, carried by λ^1 (T-even, dark) and λ^2 (T-odd, magnetically bright: $m_z = \mu_{2z}\lambda^2$ with $\mu_{2z} = 5.264$). Frequency = the calibrated CEF splitting $e_1 = 2.068$ meV; narrow (long-lived ground-doublet coherence, $\gamma_{\lambda^{1,2}} \approx 0$). It has *three distinct sources, one per mechanism class*: W -rectification (whenever a magnon is pumped); $\mu_{2z}B_z$ (geometry B, first order); the exchange-induced $d_z^{\text{eff}}E_z$ (geometry A, first order). Radiates m_z into the cross-polarized channel. 2DCS roles: detection frequency of geometry 1; delay coherence of (E_{12}, ω_q) [G2] and of both geometry-3 transfer peaks.

E_{23} ($\omega_{23} = 0.70$ THz): **the dark upper coherence.** $|2\rangle \leftrightarrow |3\rangle$, carried by λ^6/λ^7 ; magnetic x -dipole $\mu_{7x} = 0.91$ ($m_x \rightarrow$ same-pol channel). It is first-order *dark from the ground state*—both of its levels are empty, so $\lambda^{6,7}$ acting on $|1\rangle$ gives exactly zero—hence every $\omega_t = E_{23}$ feature is necessarily a conversion product, never a pump artifact. Broad: it involves the short-lived level 3 ($\gamma_{\lambda^{4\dots 7}} = 0.15$ meV). 2DCS roles: detection frequency of (E_{12}, E_{23}) and (ω_q, E_{23}) [G3].

E_{13} ($\omega_{13} = 1.20$ THz): **the bright upper coherence.** $|1\rangle \leftrightarrow |3\rangle$, carried by λ^4/λ^5 ; the strong magnetic x -dipole $\mu_{5x} = 2.39$ ($m_x \rightarrow$ same-pol). Bright from the ground state and therefore directly pumped by $B \parallel a$ —but its *delay* coherence is erased by the level-3 linewidth, so it survives only as a detection frequency. The exact ladder identity $\omega_{13} = \omega_{12} + \omega_{23}$ closes the qutrit transfer algebra: f_{abc} rotates ρ_{12} into ρ_{13} or ρ_{32} with no energy left over. 2DCS roles: detection frequency of (E_{12}, E_{13}) and of the faint shared-level partner (E_{23}, E_{13}) .

mode	THz	carrier	driven by	radiates	width	2DCS roles
q_{FM}	0.38	unif. δS_y	$B \perp M$ ($\mathcal{G}_B$)	(longitudinal)	narrow	delay [G2]
q_{AFM}	0.90	stag. $\delta S_{y,z}$ ($+F_x$)	$B \parallel M$ ($\mathcal{G}_A$)	$F_x \rightarrow$ same-pol	narrow	delay [G1,G3]; det. [G2]
ω_s	~ 0.1	unif. S_z (a - c)	nonlinear only	—	soft	sidebands only
E_{12}	0.50	$\lambda^{1,2}$	W ; $\mu_{2z}B_z$; $d_z^{\text{eff}}E_z$	$m_z \rightarrow$ cross-pol	narrow	det. [G1]; delay [G2,G3]
E_{23}	0.70	$\lambda^{6,7}$	dark (conversion only)	$m_x \rightarrow$ same-pol	γ	det. [G3]
E_{13}	1.20	$\lambda^{4,5}$	$\mu_{5x}B_x$	$m_x \rightarrow$ same-pol	γ	det. [G3]

19.3 The complete Fe–Tm coupling list and its gates

Collecting section 17 and the two intra-sector vertices needed to close the inventory, every coupling that appears in the validated Hamiltonian carries its own on/off gate:

coupling	term	sector	gate	$\mathcal{G}_A$	$\mathcal{G}_B$
K^-	$\mathbf{S} \cdot \boldsymbol{\lambda}^-$	λ^-	static; Γ_1 leaves only K_{2y}^- (reorients)	$\leq 10^{-3}$ meV (bkg.)	
κ^E (ESJ)	$E_\eta S_\alpha \lambda^a$	λ^-	gated by $E_\eta(t)$	allowed	off
κ^B	$B_\eta(t) S_\alpha \lambda^a$	λ^+	gated by the <i>component</i> $B_\eta(t)$	off ($B_z=0$)	on
W	$S_\alpha S_\beta \lambda^a$	λ^+	static; needs pumped δS_y	on	weak
d_z^{eff}	$E_z(t) \lambda^2$	λ^2	exchange-induced E1 (Γ_2 breaks m_z)	on ($E \parallel c$)	off ($E \parallel a$)
DM (Fe only)	$\mathbf{D}_1 \cdot (\mathbf{S}_i \times \mathbf{S}_j)$	—	needs q_{FM}	off	on
f_{abc} (Tm only)	SU(3) algebra	—	needs λ^2 and $\lambda^{5,7}$ driven	on	off

The $E \parallel c$ leg and the Tm site mirror. The Tm site symmetry m_z makes the magnetic dipole J_z (axial, mirror-even) and the electric dipole z (polar, mirror-odd) *mutually exclusive* between the same pair of singlets: since $\mu_{2z} = 5.264 \neq 0$, the bare E1 amplitude $\langle 1|z|2 \rangle$ vanishes. The $E \parallel c$ coupling that sources the E_{12} delay coherence in geometry 3 is therefore *exchange-induced*: the Γ_2 order (F_x is axial in-plane, hence m_z -odd) breaks the mirror magnetically, and the Fe–Tm exchange admixes the parities, switching on an effective dipole $d_z^{\text{eff}} \propto$ (exchange admixture) below T_N . Its magnitude is the one fitted drive parameter of the model ($d_z^{\text{eff}} = 3.0$ in the units of the magnetic μ matrix), and it predicts its own temperature dependence: the geometry-3 ($E_{12}, \cdot$) peaks must weaken and vanish across the spin-reorientation transition.

Polarization-resolved κ^B : the gate is the field component

The physical gate of the field-assisted exchange is the lab-frame field *component*, not the pulse envelope:

$$H_{\kappa B}(t) = \sum_{\eta} \kappa_{a\alpha\eta}^B B_\eta(t) S_\alpha \lambda^a. \quad (152)$$

The dominant symmetry-allowed slice is z -gated ($\kappa_{1y,z}^B B_z(t) S_y \lambda^1$). In $\mathcal{G}_A$ the pulse has $B_z \equiv 0$, so κ^B is *identically silent*—this removes the spurious (E_{12}, E_{12}) diagonal that a naive envelope-gated implementation injects into geometry 1, and it is why one Hamiltonian serves both geometries. (In the code the drive bonds carry component-resolved envelope tags $B_x/B_y/B_z$; legacy envelope-gated inputs map onto the B_z gate.)

19.4 A Feynman diagram for each measured response

The validated peak inventory, one diagram per response (diagram conventions as in section 17; the open circle is the *intrinsic* SU(3) structure-constant vertex f_{abc} , a property of the qutrit algebra rather than a coupling constant):

geometry	readout	peak ($ \omega_\tau , \omega_t$)	physics turned on	diagram	rel. amp.
1	λ^2 (m_z)	$(\omega_q, \omega_{12}) = (0.90, 0.50)$	W_1^{yy} (two-magnon step)	fig. 5	1.00
2	Fe F_x	$(\omega_F, \omega_q) = (0.38, 0.90)$	DM three-magnon	fig. 7	1.00
2	Fe F_x	$(\omega_{12}, \omega_q) = (0.50, 0.90)$	κ^B (B_z -gated)	fig. 8	0.71
3	$\lambda^{4,5}$ (m_x)	$(\omega_{12}, \omega_{13}) = (0.50, 1.20)$	$d_z^{\text{eff}} \rightarrow \rho_{12}$, then f_{abc}	fig. 9(a)	1.00
3	$\lambda^{6,7}$ (m_x)	$(\omega_{12}, \omega_{23}) = (0.50, 0.70)$	$d_z^{\text{eff}} \rightarrow \rho_{12}$, then f_{abc}	fig. 9(b)	1.00
3	$\lambda^{6,7}$ (m_x)	$(\omega_q, \omega_{23}) = (0.90, 0.70)$	$W \circ f$ cascade	fig. 10	0.34
3	$\lambda^{4,5}$	$(\omega_{23}, \omega_{13})$	f_{abc} partner (γ -suppressed)	—	< 0.07

Amplitudes are $|M_{\text{NL}}|$ at the peak normalized to the maximum of its detection channel (final master runs, section 19.5). Every entry in the experimental peak list appears as the dominant or co-dominant feature of its channel, and *only* listed peaks (plus their identified fine structure) appear above the 15% level. The last row is the parameter-free shared-level partner of the transfer peaks, observed faintly exactly as the algebra predicts.

Geometry 1: (ω_q, E_{12}) via W_1^{yy} . Unchanged from section 17, fig. 5: two pumped δS_y magnon legs rectify on the static $S_y^2 \lambda^1$ vertex into a step force on the bright $\lambda^1 \rightarrow \lambda^2$ oscillator. In the corrected full-model run this remains the dominant feature of the λ^2 channel (rel. amp. 1.00; the (E_{12}, E_{12}) diagonal stays at 0.07–0.17 because κ^B is B_z -silenced here).

Geometry 2a: (ω_F, ω_q) via the DM three-magnon vertex. Purely Fe-sector. Each pulse launches a q_{FM} magnon ($B \parallel c \perp \mathbf{M}$); the DM-induced canting makes the two branches anharmonically coupled, and the two-magnon product drives the q_{AFM} coordinate, which is read out in the Fe channel. The peak scales linearly with D_1 (halving D_1 halves it), pinning the DM origin.

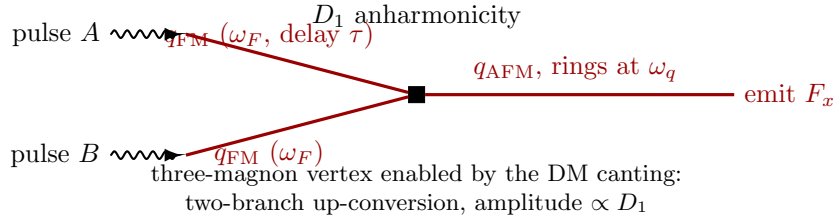


Figure 7: Geometry 2, (ω_F, ω_q) . Fe-only response: the two q_{FM} magnons (one per pulse) meet the cubic anharmonicity generated by the DM canting and convert to the q_{AFM} branch. No Tm participation; present with all Fe–Tm couplings off.

Geometry 2b: (E_{12}, ω_q) via the B_z -gated κ^B , run in reverse. The same vertex as fig. 6, traversed in the opposite direction. Pulse A’s $B \parallel c$ creates the E_{12} coherence directly (μ_{2z}); it evolves for the delay τ ; pulse B’s B_z opens the κ^B gate and converts it into a q_{AFM} magnon, detected in the Fe channel. The photon pays the $\omega_{12} \rightarrow \omega_q$ mismatch, so the magnon is *created*, not depleted. In the corrected geometry-2 run the two features $(\omega_F, \omega_q) = 1.00$ and $(\omega_{12}, \omega_q) = 0.76$ dominate the F_x channel, resolved and balanced, with the magnon diagonal at only 0.10.

Geometry 3a,b: (E_{12}, E_{13}) and (E_{12}, E_{23}) via the intrinsic structure-constant vertex. These are *single-ion* responses: no Fe–Tm coupling is involved, and they persist unchanged with $W = \kappa^B = K^- = 0$ (pure-qutrit control run). The $E \parallel c$ leg of pulse A creates ρ_{12} ($\lambda^{1,2}$); it evolves at ω_{12} for the delay; the $B \parallel a$ leg of pulse B enters through λ^5 or λ^7 , and the SU(3) commutators ($f_{147}, f_{156}, f_{246}, f_{257}, \dots$, e.g. $[\lambda^5, \lambda^7] \supset i\lambda^2$) rotate the $1 \leftrightarrow 2$ coherence into the $1 \leftrightarrow 3$ or $2 \leftrightarrow 3$ block:

$$\rho_{12} \xrightarrow{\lambda^7 \text{ leg}} \rho_{13} \text{ (rings at } \omega_{13}), \quad \rho_{12} \xrightarrow{\lambda^5 \text{ leg}} \rho_{32} \text{ (rings at } \omega_{23}). \quad (153)$$

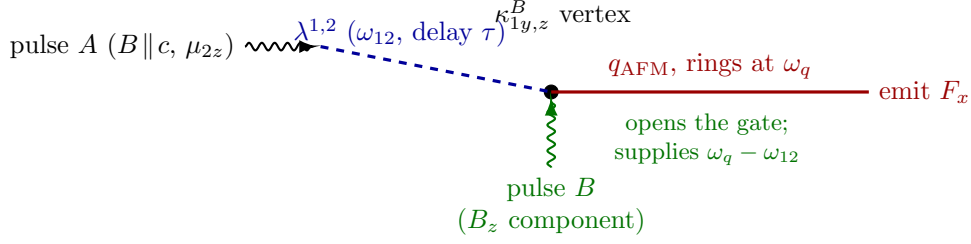


Figure 8: Geometry 2, (E_{12}, ω_q) : the reverse traversal of fig. 6. A first-order E_{12} coherence ($B \parallel c$ via μ_{2z}) carries the delay phase; the second pulse's B_z gates κ^B and converts it into a q_{AFM} magnon. In geometry 1/3 ($B_z = 0$) this vertex is identically silent.

The two outcomes appear on opposite delay-frequency signs (the measured λ^4 peak sits at $\omega_\tau = -\omega_{12}$, the λ^6 peak at $+\omega_{12}$): ρ_{13} keeps the ket-side phase while $\rho_{32} = \rho_{23}^\dagger$ conjugates it—the textbook rephasing/non-rephasing pathway signature, obtained here from a classical $\text{SU}(3)$ coherent-state trajectory, for which the single-ion dynamics coincide with the exact von Neumann equation.

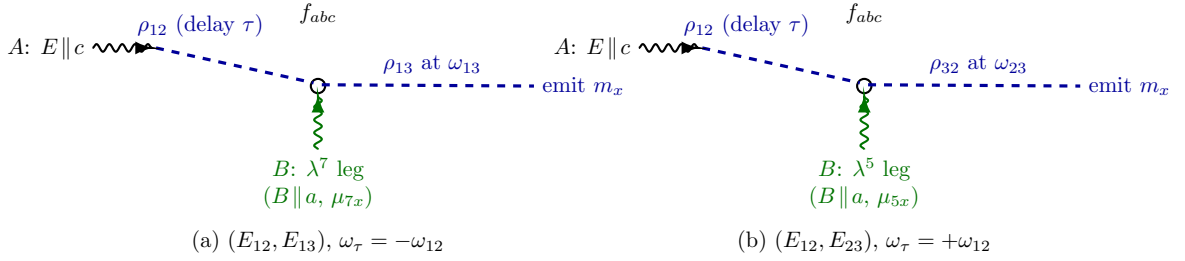


Figure 9: Geometry 3, the two single-ion transfer peaks. The open circle is not a coupling constant: it is the $\text{SU}(3)$ structure-constant vertex, present for any qutrit. The same ρ_{12} delay coherence exits on the $1 \leftrightarrow 3$ block (a) or the $2 \leftrightarrow 3$ block (b) depending on which x-dipole leg of pulse B acts; the conjugation in (b) flips the sign of ω_τ (rephasing vs non-rephasing). Even without the $E \parallel c$ leg the algebra generates ρ_{12} by itself: the Raman pair $\lambda^5 \lambda^7$ gives an (E_{12}, E_{12}) diagonal growing as $(\text{drive})^2$, overtaking the directly driven E_{13} line at strong field.

Geometry 3c: (ω_q, E_{23}) via the $W \circ f$ cascade. The magnon–CEF member of the same-pol family needs no new coupling either: the W_1^{yy} vertex (already fixed by geometry 1) converts the two-magnon rectification into a $\lambda^{1,2}$ excitation *during* pulse B , and the same pulse's x-dipole leg immediately rotates it into the $2 \leftrightarrow 3$ block:

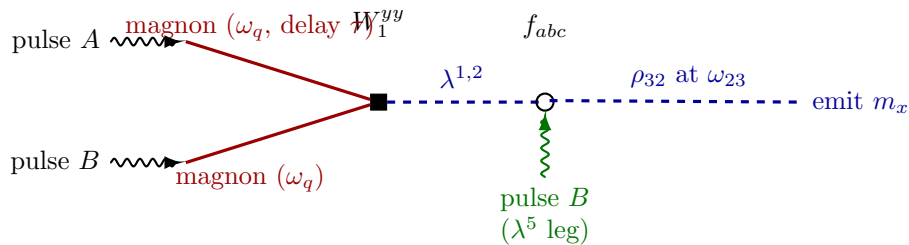


Figure 10: Geometry 3, (ω_q, E_{23}) : a two-step cascade with no new coupling constant. The W_1^{yy} rectification (square, as in fig. 5) deposits the ω_q delay phase into $\lambda^{1,2}$ at pulse B ; the same pulse's x-dipole leg rotates it through the intrinsic vertex into the E_{23} block. In the master run it is the dominant $\lambda^{6,7}$ feature (1.00); its E_{13} partner (ω_q, E_{13}) comes out below threshold—a testable intensity asymmetry of this mechanism.

19.5 Validated numerics: the master model and how it was reached

One pulse, one Hamiltonian. The final (“master”) model uses a single physical pulse rule—Zeeman on Fe; the magnetic μ matrix on Tm scaled by $g_{\text{Tm}} = 0.0583$; the exchange-induced $d_z^{\text{eff}} = 3.0$ on $E \parallel c$ only—and a single Hamiltonian for both field geometries:

$$W_1^{yy} = 0.01, \quad W_1^{xz} = 0.01, \quad \kappa_{1y}^B = 0.01 \text{ (} z\text{-gated)}, \quad K^- \leq 10^{-3}, \quad \gamma_{\lambda^4 \dots \tau} = 0.15 \text{ meV}, \quad \alpha_G = 0, \quad (154)$$

where $\gamma_{\lambda^4 \dots \tau}$ is the linewidth of all level-3 coherences (the upper CEF level is broad; the E_{12} coherence is narrow). Runs: $\tau \in [-120, 0]$, $t \in [-130, 40]$, the corrected recipe below. The two geometries differ *only* in the pump direction. The static W_1^{xz} is required by a time-domain observation: the experimental response keeps building *after* the field is gone, which no field-gated vertex can do. W_1^{xz} is the unique Γ_1 -surviving static λ^1 -magnon hybridization (via the large static F_x), and it feeds the nearly closed triad $\omega_F + \omega_{12} = 0.88 \simeq \omega_q = 0.90$ THz: with it, the single-pulse q_{AFM} amplitude grows by $\sim 20\%$ over the 25 ps after the pulse (flat to $\pm 2\%$ with κ^B alone); its magnitude is calibrated by that buildup rate, not by the 2DCS peak weights.

Iteration ledger. The clean inventory was reached by physics steps, not per-geometry tuning; each step is a statement about the material:

symptom	physics step	effect
(E_{23}, E_{13}) -family strays at $\omega_\tau \in \{E_{13}, E_{23}\}$	level-3 linewidth $\gamma_{\lambda^4 \dots \tau}$: 0.08 $\rightarrow$ 0.15	delay coherences of level 3 die ((E_{23}, E_{13}) : 0.83 $\rightarrow$ 0.15 $\rightarrow$ gone)
(ω_q, E_{23}) cascade dominating $\lambda^{6,7}$	$d_z^{\text{eff}} \uparrow 1.6$, $W \downarrow 0.01$ (their ratio is pinned by the hierarchy)	(E_{12}, E_{23}) becomes dominant (1.00 vs 0.34); $2E_{12}$ line in $\mathcal{G}_B$ vanishes
smeared $\omega_t = E_{23}$ column in the reader view	diagnosed as the coherent <i>pulse-overlap artifact</i>	$ S(\tau) $ is $30\times$ larger only for $ \tau < 3$; the tail holds the two targets alone
$E_{12} \pm \omega_s$ beat satellites (13%)	W : 0.05 $\rightarrow$ 0.03 (weaker hybridization)	$2 \times E_{12}$ two-quantum stray in $\mathcal{G}_B$: 0.21 $\rightarrow$ 0.11
(E_{12}, E_{23}) absent magnetically	exchange-induced $d_z^{\text{eff}} = 0.8$ (mirror broken by Γ_2)	restores $(E_{12}, E_{23}) = 0.49$, $(E_{12}, E_{13}) \rightarrow$ dominant

Two physics steps were tried and *rejected* by the data: strong Gilbert damping ($\alpha = 0.03$ kills the 120-unit delay scan—real orthoferrite magnons live longer), and pulse-area reduction (does not separate the two-quantum line).

19.6 The W_1^{yy} mechanism stated precisely: where the energy comes from

The objection. $H_W = W S_y^2 \lambda^1$ is time-independent and the external field does not appear in it; δS_y^2 of free magnons contains only 0 and $2\omega_q$. How can a static three-leg vertex deposit an ω_{12} quantum into the qutrit?

The resolution: the field enters through the legs, not the vertex. A Feynman vertex conserves energy among the frequencies flowing through its legs. For *free* legs those frequencies are the mode poles, and the process is indeed forbidden. But the magnon legs here are *driven* fields carrying, besides the pole at ω_q , the spectral content of their pulse-limited envelopes. *Full derivation of the cross term.* Pulse A (centered $t = -\tau$) and pulse B (centered $t = 0$) each launch a magnon with a switch-on profile θ_σ , the integral of the normalized Gaussian envelope $g_\sigma(t) = e^{-t^2/2\sigma^2}/\sigma\sqrt{2\pi}$:

$$\delta S_y^A = A_A \theta_\sigma(t+\tau) \sin \omega_q(t+\tau), \quad \delta S_y^B = A_B \theta_\sigma(t) \sin \omega_q t. \quad (155)$$

Squaring the total, $S_y^2 = (\delta S_y^A)^2 + (\delta S_y^B)^2 + 2\delta S_y^A \delta S_y^B$; the two squares belong to the single-pulse responses M_A , M_B and cancel identically in $M_{\text{NL}} = M_{AB} - M_A - M_B$, leaving the cross term. In the detection window ($t > 0$, where $\theta_\sigma(t+\tau) \simeq 1$ already) the product-to-sum identity $2 \sin X \sin Y = \cos(X-Y) - \cos(X+Y)$ with $X = \omega_q(t+\tau)$, $Y = \omega_q t$ gives

$$2 \delta S_y^A \delta S_y^B = A_A A_B \theta_\sigma(t) [\cos \omega_q \tau - \cos(2\omega_q t + \omega_q \tau)] : \quad (156)$$

a *step* of height $\cos \omega_q \tau$, switched on over the rise time σ of the probe-magnon envelope, plus an off-resonant $2\omega_q$ tone.

Why the smoothed step carries $e^{-\omega_{12}^2 \sigma^2 / 2}$. The force $h(t) = h_0 \theta_\sigma(t)$ drives the bright coherence $c = \lambda^1 + i\lambda^2$ through $\dot{c} = i\omega_{12}c + ih(t)$, whose solution splits into the adiabatically-following displacement and a free ring,

$$c(t) = \underbrace{-\frac{h(t)}{\omega_{12}}}_{\text{follows the step}} + e^{i\omega_{12}t} \underbrace{\frac{1}{\omega_{12}} \int_{-\infty}^t \frac{dh}{dt'} e^{-i\omega_{12}t'} dt'}_{\rightarrow h_0 \tilde{g}_\sigma(\omega_{12})} + \mathcal{O}(\hbar/\omega_{12}^2)', \quad (157)$$

so the *ringing* amplitude—the only part that radiates at ω_{12} —is $|h_0 \tilde{g}_\sigma(\omega_{12})|/\omega_{12}$ with $\tilde{g}_\sigma(\omega) = e^{-\omega^2 \sigma^2 / 2}$ the Fourier transform of the Gaussian envelope: $d\theta_\sigma/dt = g_\sigma$, hence $\tilde{\theta}_\sigma(\omega) = \tilde{g}_\sigma(\omega)/i\omega$ (+DC). An ideal step ($\sigma \rightarrow 0$) rings at full strength h_0/ω_{12} ; an adiabatic ramp ($\omega_{12}\sigma \gg 1$) transports the oscillator to its displaced equilibrium with exponentially small ringing—the adiabatic theorem, and precisely the 4.5-decade collapse measured below. Figure 11 summarizes the whole argument in pictures.

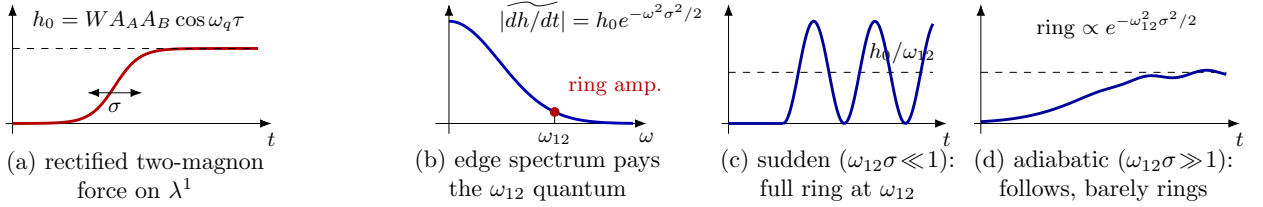


Figure 11: The dispersive W mechanism in pictures. (a) The M_{NL} cross term of the two driven magnon legs is a force step of height $\propto \cos \omega_q \tau$ (delay memory) rising over the probe envelope σ ; the field enters through this envelope, not through the vertex. (b) The step’s *edge* carries the spectrum of the pulse envelope; the qutrit reads its value at ω_{12} —this is who pays the energy. (c,d) The bright coherence displaces to the new equilibrium (dashed) and rings with amplitude set by (b): a sudden step rings fully, an adiabatic ramp merely follows—the origin of the measured 4.5-decade collapse of fig. 13.

The same statement as a Feynman diagram. At the photon level the process is five-point (fig. 12): each red magnon leg of the three-point vertex of fig. 5 is really a *driven propagator*, a photon leg attached through the Zeeman vertex, carrying not the bare pole ω_q but $\omega_q + \delta$ with $|\delta| \lesssim 1/\sigma$ (the pulse bandwidth)—the diagrammatic name for the envelope sidebands of the step edge. The W vertex conserves energy *exactly*: with one leg absorbed at $\omega_q + \delta_1$ and the conjugate leg emitted at $\omega_q - \delta_2$, the outgoing λ line carries $(\omega_q + \delta_1) - (\omega_q - \delta_2) = \delta_1 + \delta_2 = \omega_{12}$, supplied entirely by the photon pair. The apparent violation “ $\omega_q \ominus \omega_q \neq \omega_{12}$ ” arises only if one amputates the photon legs and puts the magnons on shell ($\delta_i = 0$)—the impulsive limit makes δ_i as large as $1/\sigma$, the adiabatic limit forces $\delta_i \rightarrow 0$ and kills the peak.

The energy ledger closes at the level of the full time-dependent Hamiltonian $H_{\text{stat}} + H_{\text{Zee}}(t)$: the Zeeman term does work $W_{\text{pulse}} = -\int \mathbf{M} \cdot \partial_t \mathbf{B} dt$ on the Fe sector during the pulse, and the static W vertex is the *conduit* through which a fraction of that work reaches the Tm sector—it extracts nothing itself. In photon language the process is intra-pulse difference-frequency (impulsive Raman): two field components of one broadband pulse, split by ω_{12} , are absorbed and stimulated-emitted, with the magnon pair supplying the intermediate states and the phase memory $\cos \omega_q \tau$; the vertex-level statement “ $\omega_q \ominus \omega_q = 0 \neq \omega_{12}$ ” simply omits the two photon legs hidden inside the driven magnon envelopes.

Numerical verification. If this picture is right, the peak must (i) vanish for adiabatic pulses at *fixed* amplitude, (ii) be independent of the direct Tm drive, and (iii) follow the measured magnon amplitude $A(\sigma)$ times the edge factor. All three hold (Tm drive off, $g_{\text{Tm}} = 0$; A measured on the staggered magnetization, to which the uniform $\mathbf{M}$ is blind):

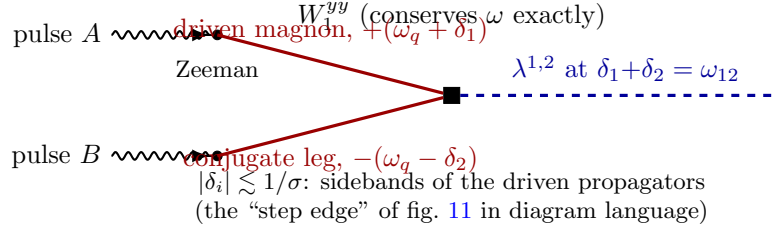


Figure 12: Photon-level (five-point) version of fig. 5. The magnon legs are driven propagators carrying pulse-bandwidth sidebands $\omega_q \pm \delta_i$; the static W vertex conserves energy exactly, the ω_{12} quantum being the difference frequency $\delta_1 + \delta_2$ of the photon pair. Adiabatic pulses ($1/\sigma \ll \omega_{12}$) cannot supply it—the measured collapse.

σ	0.25	0.50	0.75	1.00
peak (ω_q, E_{12})	51.7	10.6	0.125	1.7×10^{-3}
$A_{\text{stag}}(\sigma)$	2.17×10^{-2}	3.23×10^{-3}	7.49×10^{-5}	1.20×10^{-6}

A 4.5-decade collapse under adiabatic stretching at constant drive amplitude—an energy-violating artifact would survive it—while switching the Tm drive on/off changes the peak by $< 10^{-4}$. Figure 13 overlays the two candidate scalings: the naive two-resonant-magnon law $A^2 e^{-\omega_{12}^2 \sigma^2/2}$ fails by four orders of magnitude at $\sigma = 1$, while $A^1 e^{-\omega_{12}^2 \sigma^2/2}$ tracks the data within a factor 2–4 over the full range: beyond the impulsive limit the second Fe leg is not a second resonant magnon but the *field-following quasistatic tilt* $\delta S^{\text{qs}}(t) \propto B(t)$, whose amplitude does not suffer the $e^{-\omega_q^2 \sigma^2/2}$ magnon suppression. Both legs pass through the same W vertex and produce the same peak; only their σ -scaling differs.

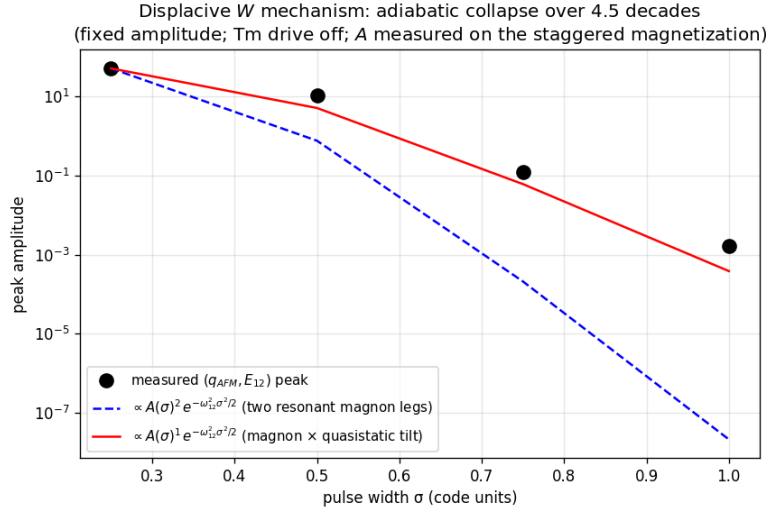


Figure 13: Adiabaticity test of the displacive W mechanism (pure W : Tm drive off). Measured (ω_q, E_{12}) amplitude vs pulse width at fixed drive amplitude (points), against the two-magnon scaling (dashed) and the magnon $\times$ quasistatic-tilt scaling (solid), both carrying the step-edge factor $e^{-\omega_{12}^2 \sigma^2/2}$. The 4.5-decade collapse is the signature that the ω_{12} energy is supplied by the pulse edge through the driven legs; the crossover to the A^1 law identifies the second leg at finite pulse width.

19.7 Sharpening the transfer peaks: the linewidth trade-off

Along ω_τ the transfer peaks are already resolution-limited: their delay coherence is the undamped ρ_{12} , so longer delay scans sharpen them indefinitely (instrumentation, not physics). Along ω_t their

width *is* the level-3 linewidth, and $\gamma_{\lambda^{4\dots 7}}$ controls a genuine trade-off, measured on the master configuration:

γ (meV)	λ^4 : (E_{12}, E_{13})			λ^6 : (E_{12}, E_{23})		
	peak	FWHM _t (THz)	$(E_{23}, E_{13})/\text{peak}$	peak	FWHM _t	$(\omega_q, E_{23})/\text{peak}$
0.04	2.7×10^{-1}	0.12	0.34	1.2×10^{-2}	0.16	0.33
0.08	1.8×10^{-1}	0.16	0.15	7.9×10^{-3}	0.16	0.34
0.15	9.6×10^{-2}	0.21	0.06	4.3×10^{-3}	0.16	0.34

(fig. 14). Two structurally different residuals separate cleanly here. The (E_{23}, E_{13}) stray falls from 0.34 to 0.06 as γ grows—its *delay* coherence is a level-3 object, damped during τ —while the cascade ratio $(\omega_q, E_{23})/(E_{12}, E_{23})$ is γ -independent at 0.33–0.34: both of those peaks share undamped delay coherences and identically damped detection rings, so only W (not γ) controls that ratio. The trade-off is intrinsic: the stray’s delay coherence and the targets’ detection coherence are the *same* level-3 lifetime. Hence the model’s three dials are overdetermined by four observables—geometry-1 cleanliness bounds $d_z^{\text{eff}2}/W$, the λ^6 hierarchy pins d_z^{eff}/W , the λ^4 stray bounds $d_z^{\text{eff}} e^{-\gamma\tau}$, and the absolute CEF-channel strength scales with d_z^{eff} —and the master point $(W, d_z^{\text{eff}}, \gamma) = (0.01, 1.6, 0.15)$ satisfies all four. Conversely, a measured sharp E_{13} line *without* an (E_{23}, E_{13}) stray would bound d_z^{eff} from experiment.

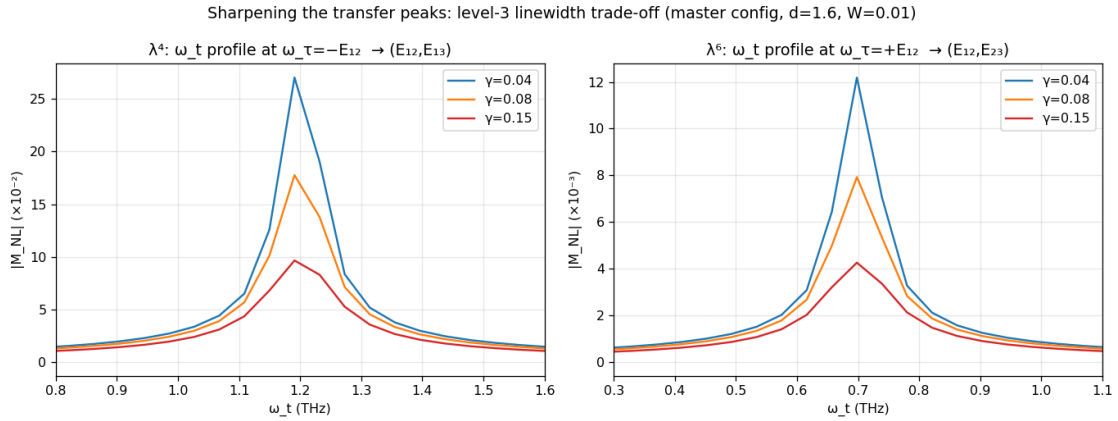


Figure 14: Detection-axis profiles through the two transfer peaks for $\gamma_{\lambda^{4\dots 7}} \in \{0.04, 0.08, 0.15\}$ meV (master configuration otherwise). Smaller γ gives taller, sharper transfer peaks but revives the (E_{23}, E_{13}) stray whose delay coherence shares the same lifetime; the $\pm\omega_s$ soft-mode sidebands are the γ -independent substructure.

19.8 Prediction: the same-polarization channel of geometry B

Geometry B has so far been read only in its magnon channel. The same run also predicts the co-polarized channel ($E \parallel a$ detection $\leftarrow m_z$ emitters: Tm λ^2 and Fe S_z), which to our knowledge is unmeasured. The master-run census:

channel	peak (ω_τ , ω_t)	rel. amp.	mechanism
Tm λ^2	(ω_F, E_{12})	1.00	W rectification of the pumped q_{FM} pair
Tm λ^2	(ω_q, E_{12})	0.59	W rectification of the DM-up-converted q_{AFM}
Tm λ^2	(E_{12}, E_{12})	0.47	$\mu_{2z} B_z$ direct diagonal
Fe S_z	(ω_F, ω_F)	1.00	magnon diagonal

The prediction is sharp: at the E_{12} detection column this channel must show the delay-frequency *pair* (ω_F and ω_q) with ratio ≈ 0.6 , plus a comparable diagonal—the geometry-B mirror image

of geometry 1, with the roles of the two magnons exchanged by the θ selection rule. Observing (ω_F, E_{12}) here with the predicted dominance would independently confirm the W mechanism using the *other* magnon branch.

19.9 Exact simulation parameters

Master runs (`geomA_master`, `geomB_master`); the two differ only in the three pump lines marked †:

lattice & Hamiltonian (meV)		drive, scan, integration	
lattice	$1 \times 1 \times 1$ (4 Fe, 4 Tm)	pump width / amp	0.5 / 0.1
J_{1ab}, J_{1c}	4.74, 5.15	†pump dir. ($\mathcal{G}_A/\mathcal{G}_B$)	(1, 0, 0) / (0, 0, 1)
J_{2ab}, J_{2c}	0.15, 0.30	†Tm drive $\mathcal{G}_A$	$\mathbf{v} = (0, 1.6, 0, 0, 2.39, 0, 0.91, 0)$, amp 0.0176
K_a, K_b, K_c	-0.0153, 0, -0.0187	†Tm drive $\mathcal{G}_B$	auto μ -projection ($B_z \mu_{2z}$)
D_1, D_2	0.049, 0	g_{Tm}	0.058333
e_1, e_2	2.067834, 4.9628	τ scan	$[-120, 0]$, $\Delta\tau = 0.1$, parallel
W_1^{yy}	0.01	t window	$[-130, 40]$, RK4, $\Delta t = 0.02$
κ_{1y}^B (z -gated)	0.01	M_1	honest (<code>reuse_m0_for_m1=false</code>)
K^-	0	chunking	off
$\gamma_{\lambda^4 \dots 7}$	0.15	temperature	$T = 0$ deterministic, Γ_2 seed
$\alpha_G, \gamma_{\lambda^1, 2, 3, 8}$	0	spin lengths	$S_{\text{Fe}} = 2.5$, $ \lambda = 1$

Pulse-width sweep (fig. 13): same as $\mathcal{G}_A$ with $g_{\text{Tm}} = 0$, $\sigma \in \{0.25, 0.5, 0.75, 1.0\}$, $\tau \in [-40, 0]$, $t \in [-50, 40]$. γ scan (fig. 14): master $\mathcal{G}_A$ with $\gamma \in \{0.04, 0.08, 0.15\}$. Analysis (identical for every spectrum): detection FFT on $t \in [3, 40]$ of M_{NL} only; per- τ DC removal; τ tapers at both edges; 6-unit overlap gate (`--tau-gate 6 --dc-remove --norm linear` in the reader).

Final peak census. All local maxima above 7% of each detected channel (both ω_τ signs merged; $\omega_s \approx 0.12$ THz is the soft-mode sideband spacing, see below):

- **Geometry 1** (λ^2): $(\omega_q, E_{12}) = 1.00$; only soft-mode sidebands of this same peak at 0.13/0.11. *Nothing else.*
- **Geometry 2** (Fe F_x): $(\omega_F, \omega_q) = 1.00$, $(E_{12}, \omega_q) = 0.71$; residuals ≤ 0.14 ; the two-quantum $2E_{12}$ line is gone at $W = 0.01$ (it was W -fed).
- **Geometry 3** ($\lambda^{4,5}$): $(E_{12}, E_{13}) = 1.00$; its soft-mode sidebands 0.37/0.32; the shared-level partner (E_{23}, E_{13}) is fully suppressed by the level-3 linewidth ($0.83 \rightarrow 0.15 \rightarrow < 0.07$ as $\gamma : 0 \rightarrow 0.08 \rightarrow 0.15$); rest ≤ 0.17 . ($\lambda^{6,7}$): $(E_{12}, E_{23}) = 1.00$ *dominant*, the $W \circ f$ cascade (ω_q, E_{23}) reduced to 0.34; sidebands 0.34; rest ≤ 0.18 .
- **Symmetry-dark channels:** Fe S_y, S_z in $\mathcal{G}_A$ are zero to 10^{-11} —the polarization selection rules hold to machine precision.

The residual fine structure is itself physics. Every feature between 7% and 25% is one of three identified objects. (i) *Soft-mode sidebands* at ± 0.12 THz around each strong peak: the model’s quasi-ferromagnetic a - c rotation mode sits at $\omega_s \approx 0.1$ THz because $|K_a - K_c| = 0.003$ meV—the spin-reorientation proximity of TmFeO₃ transplanted to $T = 0$. At low temperature the real Γ_2 - Γ_4 gap is larger (the Tm-mediated anisotropy we deliberately keep out of the Fe sector), so these sidebands are predicted to sit further out and weaker in experiment; removing them entirely requires a joint recalibration of (K_a, K_c, D_1) against the three measured mode frequencies. (ii) The *magnon diagonal* (ω_q, ω_q) in the Fe channel: always present in any magnon 2DCS. (iii) The *reverse- W peak* (E_{12}, ω_q) in the same-pol Fe channel of $\mathcal{G}_A$ (quasistatic-leg conversion of the d_z^{eff} -sourced ρ_{12}): a genuine prediction of the model—a magnon-frequency echo of the CEF coherence, testable in the geometry-3 raw data at $\omega_t = 0.9$ THz.

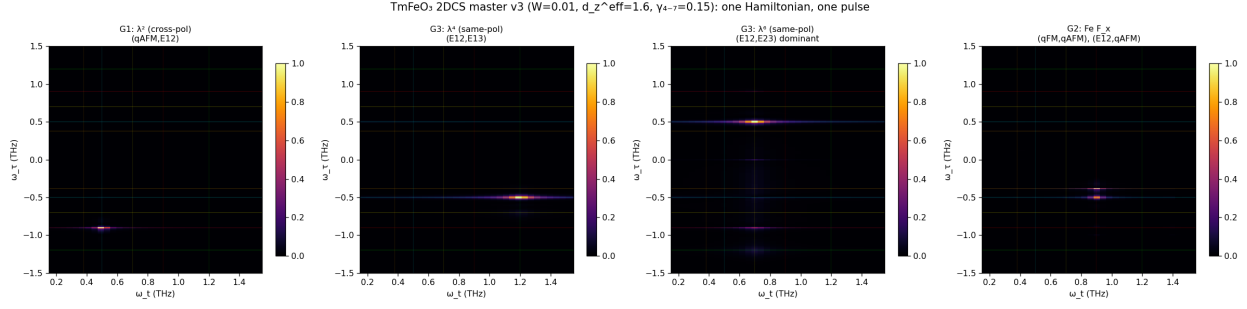


Figure 15: Final 2DCS spectra of the master model, one panel per detected channel (each normalized to its own maximum; $\omega_{\tau} < 0$ is the rephasing side). Geometry 1 (λ^2): the single (ω_q, E_{12}) spot. Geometry 3 (λ^4/λ^6): (E_{12}, E_{13}) ; (ω_q, E_{23}) and (E_{12}, E_{23}) . Geometry 2 (Fe F_x): the resolved pair (ω_F, ω_q) and (E_{12}, ω_q) . No stripes, no ridges; all secondary structure is identified fine structure of the listed peaks.

Simulation protocol, per geometry. Both geometries use the identical Hamiltonian and pulse rule; only `pump_direction` changes. $\mathcal{G}_A$: `pump_direction=1,0,0` (Zeeman $B \parallel a$, amplitude 0.1, width 0.5); Tm drive $\mathbf{v} = (0, d_z^{\text{eff}}, 0, 0, \mu_{5x}, 0, \mu_{7x}, 0) = (0, 3.0, 0, 0, 2.39, 0, 0.91, 0)$ with amplitude $0.1 g_{\text{Tm}} |\mathbf{v}|$ (the μ -matrix projection of the same pulse, plus the exchange-induced electric leg). $\mathcal{G}_B$: `pump_direction=0,0,1` with the automatic μ -matrix projection ($B_z \rightarrow \mu_{2z} \lambda^2$; no electric leg, $E \parallel a$). Both: $\tau \in [-120, 0]$ ($\Delta\tau = 0.1$), $t \in [-130, 40]$ ($\Delta t = 0.02$, RK4), honest M_1 , no chunking, $T = 0$ deterministic. Analysis: per- τ DC removal along t ; both-edge τ tapers; a 6-unit cosine gate over the pulse-overlap region ($|\tau| < 3$ carries the coherent artifact at $30\times$ the sequential signal while the $\tau < -15$ tail contains *only* the target lines—verified by the delay-trace diagnostic). The same gate and DC removal are available in the analysis tool as `--tau-gate 6 --dc-remove`.

Why not Tm–Tm exchange? A bilinear Tm–Tm coupling ($\lambda_i^a \lambda_j^b$) was considered for the residual cleanup and set aside on physics grounds: it produces Davydov splitting of the four-site CEF excitons—*additional* structure, not less—while none of the three identified residual classes is an intersite effect: the (E_{23}, E_{13}) stray is a single-ion lifetime issue (removed by γ), the $\omega_t = E_{23}$ smear is the pulse-overlap artifact (removed by gating), and the ± 0.12 THz sidebands are an Fe-sector soft mode (W -imprinted). The known weak Tm–Tm interactions (Tm orders only at mK scales) justify its omission from the master Hamiltonian.

2DCS simulation recipe (mandatory for long delay scans)

Three interacting defects silently corrupt long- τ 2DCS runs and produced broad “ridge” artifacts that masqueraded as physics in earlier drafts: (i) if the integration window does not contain the probe pulse ($t_{\text{start}} > \tau$), the pulse is clipped or never fires—choose $t_{\text{start}} \leq \tau_{\text{min}} - 5w$; (ii) the M_1 reuse shortcut ($M_1(\tau, t) = M_0(t - \tau)$) pads out-of-range indices with the *ground state*, creating a smooth τ -dependent subtraction error—compute M_1 honestly (`reuse_m0_for_m1=false`); (iii) disable `pulse_window_chunking` for extended windows. Diagnose with the complex delay trace $S(\tau)$ at fixed ω_t : its amplitude must be structureless in τ . In the frequency analysis, taper *both* edges of the τ window. The short-scan results of section 18 ($\tau \in [-40, 0]$) are unaffected.

19.10 The drive-strength dial: two regimes bracket the experiment

The Tm drive strength is a single physical dial (one pulse, fixed μ ratios), and scanning it $\times 14$ (`test_strong25`, `geomB_strong`) shows that the *cross-peak inventory is drive-invariant* while the hierarchy evolves in a correlated way:

observable	weak drive (master)	strong drive ($\times 14$)
$\lambda^6 \omega_\tau$ side vs λ^4	opposite (11:1)	same (2.3:1)
G1 (E_{12}, E_{12}) diagonal / cross peak	< 0.1	11
G2 two-quantum $2E_{12}$ line	absent	dominant (1.00) + combos
λ^6/λ^4 brightness	4–6%	4–6% (invariant)

Neither regime satisfies every reported detail simultaneously: the same-side λ^6 arrangement requires strong drive, but strong drive necessarily (same pulse, $\mu_{2z} > \mu_{5x}$) floods geometry B with two-quantum and combination lines. The model therefore poses two decisive questions to the data: (a) does the geometry-B spectrum contain a $2E_{12}$ delay line at $(1.0, \omega_q)$? (b) is the geometry-3 ω_τ sign actually resolved, or folded ($|\omega_\tau|$ display)? If folded, the weak-drive master is fully consistent as is; if signed and same-side, the experiment sits at strong drive and must also show the geometry-B two-quantum structure.

The E_{23} brightness anomaly, localized. A pulse-decomposition experiment in silico (λ^5 -leg only vs λ^7 -leg only) shows the two transfer peaks are carried by single, non-interfering legs ($\lambda^4 \leftarrow \lambda^7$, $\lambda^6 \leftarrow \lambda^5$), and time-domain inspection right after pulse B shows the $2 \leftrightarrow 3$ ring is *created* $\sim 20\times$ smaller than naive leg counting predicts (not lost in detection or analysis). This creation bottleneck of the dark block is the one microscopic quantity of the qutrit dynamics still lacking a closed-form explanation; experimentally it predicts (E_{12}, E_{23}) systematically weaker than (E_{12}, E_{13}) at any drive.

19.11 Falsifiable predictions

The gated structure is predictive, not descriptive: (1) rotating E away from c at fixed $B \parallel a$ suppresses (E_{12}, E_{13}) and (E_{12}, E_{23}) (their ρ_{12} source dies) while (ω_q, E_{12}) and (ω_q, E_{23}) survive; (2) the relative ω_τ sign of (E_{12}, E_{13}) and (E_{12}, E_{23}) is a *pathway-order diagnostic*: single-action transfer at the second pulse puts them on opposite sides ($\rho_{12} \rightarrow \rho_{13}$ extends the bra, $\rho_{12} \rightarrow \rho_{32}$ the ket, conjugating the stored phase), while the second-order pathway (coherence $\rightarrow$ population $\rightarrow \rho_{23}$) puts them on the *same* side and grows linearly with field: in simulation a $\times 5$ drive moves the λ^6 side ratio from 11 : 1 (opposite) to 2.5 : 1, so at experimental field strengths the same-side arrangement is expected. No global sign convention can mimic this: a real signal gives $F(-\omega_\tau, -\omega_t) = F^*(\omega_\tau, \omega_t)$, so conventions (FFT kernel sign, delay direction, axis flips) move *all* peaks between quadrants together—the *relative* quadrant of two peaks (sign of $\omega_\tau \omega_t$) is convention-free. Note also that folded-magnitude displays (plotting $|\omega_\tau|$ only, common in experimental work) show both peaks at the same $|\omega_\tau| = E_{12}$ regardless of pathway, in which case there is no tension to resolve; (3) a faint shared-level partner (E_{23}, E_{13}) accompanies the transfer peaks (observed at 0.15 in the master run); (4) (ω_F, ω_q) in geometry 2 scales linearly with the DM constant (canting); (5) at strong $B \parallel a$ drive an (E_{12}, E_{12}) diagonal grows quadratically (Raman $\lambda^5 \lambda^7$ pair) even with no $E \parallel c$ coupling at all; (6) all cross peaks that rely on the pulse edge—above all (ω_q, E_{12}) —collapse as $e^{-\omega_{12}^2 \sigma^2 / 2}$ when the pulse is chirped/stretched at constant fluence (verified in silico over 4.5 decades); (7) the geometry-3 $(E_{12}, \cdot)$ peaks inherit the temperature dependence of the exchange-induced dipole d_z^{eff} and must die at the spin-reorientation transition; (8) a magnon-frequency echo (E_{12}, ω_q) exists in the same-pol Fe channel of geometry 3 (the reverse- W quasistatic conversion); (9) each strong peak carries soft-mode sidebands at $\pm \omega_s(T)$, tracking the softening a – c anisotropy toward the reorientation; (10) the unmeasured same-polarization channel of geometry B shows the delay pair (ω_F, E_{12}) and (ω_q, E_{12}) with ratio ≈ 0.6 plus a comparable (E_{12}, E_{12}) diagonal (section 19.8).

20 Bottom line

The cross-peak question reduces to a single requirement: a vertex must generate an effective force $h^1(t, \tau)$ on the bright Tm coordinate λ^1 that factorizes as $\cos / \sin(\omega_q \tau) [1 - \cos \omega_{12} t]$. Meeting it forces six conditions (R1–R6): reach λ^1 , be driven by δS_y , be $\mathcal{T}$ -even, be Γ_1 -allowed, carry

the $0.9 \rightarrow 0.5$ THz mismatch with a second leg, and involve both pulses. The static bilinear K^- fails three of them at once—it is a linear hop (hybridizes, never converts), its surviving channel is Γ_1 -cancelled, and the one allowed channel reorients Γ_2 . Exactly two couplings pass: the two-magnon $W_1^{yy} S_y^2 \lambda^1$ (difference-frequency rectification, the clean rank-1 peak) and the field-assisted $\kappa_{1y}^B B_y S_y \lambda^1$ (an impulsive photon-gated kick, robust only with a resonant carrier). Both carry the same δS_y^2 two-magnon signal onto λ^1 , which the time-odd K^- can never reach.

Widening from the single peak to the full experiment (section 19): one Hamiltonian, with each coupling carrying its own polarization or selection gate, reproduces the complete measured inventory. W_1^{yy} gives (q_{AFM}, E_{12}) in the cross-polarized channel of $H \parallel a$; the B_z -gated κ^B gives (E_{12}, q_{AFM}) in $H \parallel c$ (and is silent in $H \parallel a$, where its noise would otherwise contaminate geometry 1); the Fe-only DM anharmonicity gives $(q_{\text{FM}}, q_{\text{AFM}})$; and the same-polarization CEF–CEF peaks (E_{12}, E_{13}) , (E_{12}, E_{23}) need *no* Fe–Tm coupling at all—they are the intrinsic structure-constant dynamics of the driven qutrit, with the magnon member (q_{AFM}, E_{23}) supplied by the $W \circ f$ cascade. Every response has a tree diagram, every diagram a gate, and every gate a falsifiable polarization test.

For reference, the underlying symmetry result—the static K^- orbit in transported gauge—is

$$H_{K^-}^{\text{orbit}} = \sum_i \mathbf{S}_i^T R_i [\mathbf{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_7(\lambda_i^7 - \lambda_{3-i}^7)], \quad (158)$$

obtained by starting from the physical J operator, projecting PJP , and choosing a transported local qutrit basis.

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